Arithmetic structure of the standard genetic code

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Abstract

The standard genetic code is near-universal and stable on the timescale of biological evolution, motivating study of its arithmetic properties. We combine proton and neutron counts of the encoded amino acids with two intrinsic groupings: the Rumer partition by family degeneracy and the three chemical axes of the third codon position (Keto/Amino, Strong/Weak, Purine/Pyrimidine). Setting aside the three stop codons and the ATG initiator leaves 60 codons over which we isolate properties that jointly determine an arithmetic structure governed by divisibility modulo 37. Among them, neutron and proton counts sit differently against the 37-lattice: neutrons reach it, while protons—necessarily even—fall one below its odd multiples. This holds globally (neutron count divisible by 37, proton count residue $-1 \pmod{37}$) and in the single pair tryptophan/isoleucine carrying the Keto/Amino asymmetry, with differences 37 and 36. Balance constraints between the Keto and Amino branches and this global divisibility yield a residue equation with constant term $111 = 3 \cdot 37$; with $\gcd(4, 37) = 1$, this fixes a unique minimal integer assignment for the four service codons, after which the regularities form a connected family of identities and divisibilities. Across the 27 NCBI translation tables, all of these hold simultaneously only in the standard code; deterministic controls localize the structure at the third codon position and at modulus 37; and against an explicit random-code null model, it is highly atypical. Throughout, the regularities are properties of codon-group sums, not individual amino acids, and reflect degeneracy alongside nucleon composition.

Contents

1 Introduction

The genetic code is one of the most ancient and conserved molecular features of cellular organisms. Its near-complete universality across the known domains of life, and the limited scope of variation in the known alternative codes, indicate that the basic structure of the standard code was already established at the early stages of biological evolution. Any organizational regularities present in its arithmetic structure therefore constitute one of the few available sources of information about the early stages of life, and require precise analysis.

The arithmetic structure of the standard genetic code has been studied since shortly after its decipherment. Yu. B. Rumer [3, 4] introduced the partition of the 64 codons into two classes of 32 codons each, Octet I (the eight four-fold degenerate families) and Octet II (the remaining codons), related by the involution $R = (\mathrm{T} \leftrightarrow \mathrm{G}, \mathrm{C} \leftrightarrow \mathrm{A})$. Shcherbak and Makukov [5] observed that the number 37 plays a recurrent role in the arithmetic of the code, appearing as a divisor of amino acid nucleon sums across various groupings of codons. Panov and Filatov [2] revisited this analysis, introducing the per-codon counting rule, which forms the methodological basis of the present study.

The present analysis uses the proton and neutron counts of the twenty canonical amino acids — parameters whose definition is unambiguous — together with the two grouping systems intrinsic to the code. The first is the Rumer partition. The second is given by the three chemical axes of the third codon position — Keto/Amino, Strong/Weak, and Purine/Pyrimidine — which exhaust the partitions of the four nucleotide bases into two pairs and correspond to the functional-group, hydrogen-bond, and ring-topology dichotomies of nucleobase chemistry.

The present work shows that within this framework the arithmetic regularities of the standard genetic code organize into a connected family of identities and divisibilities. The analysis traces this connected structure to a small number of empirical properties of the standard code. The standard code is then evaluated alongside the full registry of alternative NCBI translation tables on the same arithmetic criteria, and is shown to exhibit a number of structural regularities that distinguish it from all known natural alternatives. The specificity of the structure is examined further by deterministic controls, which test its dependence on the third codon position and on the modulus, and by a Monte Carlo test against an explicit random-code null model.

Two methodological points should be stated at the outset, since they determine how the results are to be read. First, the analysis is organized into two levels. The principal regularities are established in a parameter-independent representation, in which the three stop codons and the start codon carry no nucleon values and only the remaining 60 codons are summed; these regularities involve no assigned values whatsoever. A second level concerns the complete codon table, in which the four service codons must be given nucleon contributions. The values used there are not chosen to produce the regularities: they are obtained as the unique minimal nonnegative integer solution of a system of symmetry conditions (Section 3), so that the assignment is derived rather than fitted, and the divisibilities that follow are consequences of that derivation rather than inputs to it. Second, these assigned values are a formal device. A stop signal is not translated into an amino acid, and the start codon is taken here in its punctuation role as an initiation signal rather than as an internal methionine; attributing (P, N) values to these service codons is a means of rendering the arithmetic of the complete table visible as a single object, not a claim about any physical molecule. Sums that include service codons under such an assignment are formal integer quantities; the structure they make visible is fixed by the symmetry conditions independently of which particular solution is chosen, the assignment used here being the minimal such solution.

2 Methods

This section defines the nucleon parameters, the codon group systematics, and the parametrizations of service codons used throughout the analysis, together with the computed datasets supporting it.

2.1 Nucleon composition of amino acids

Throughout this work, each of the twenty canonical amino acids is represented as a neutral free amino acid molecule (not as a residue in a peptide chain), with its molecular formula taken from standard biochemical references. From this formula we compute four integer quantities that characterize the nucleon content of the molecule.

The proton count P is defined as the sum of atomic numbers over all atoms in the molecular formula:

$$P = \sum_i Z_i,$$

with $Z(\mathrm{H}) = 1$, $Z(\mathrm{C}) = 6$, $Z(\mathrm{N}) = 7$, $Z(\mathrm{O}) = 8$, $Z(\mathrm{S}) = 16$.

The neutron count N is computed under the assumption that every atom is present as its most abundant stable isotope. Under this convention we use $^{1}\mathrm{H}$, $^{12}\mathrm{C}$, $^{14}\mathrm{N}$, $^{16}\mathrm{O}$ and $^{32}\mathrm{S}$, so that for each atom the neutron contribution is $A_i - Z_i$, where A_i is the mass number of the chosen isotope. The total neutron count of the molecule is

$$N = \sum_i (A_i - Z_i).$$

The total nucleon count T and the proton–neutron difference Δ are defined as

$$T = P + N, \qquad \Delta = P - N.$$

Both quantities are integers by construction. Under the isotope assumptions above, hydrogen is the only element contributing to Δ, since all other isotopes used have equal numbers of protons and neutrons. Consequently Δ is numerically equal to the number of hydrogen atoms in the molecule, and in particular $\Delta \geq 0$ for every amino acid.

To avoid notational collision, we use the uppercase symbol Δ exclusively for the proton–neutron imbalance of a single molecule or codon group, and reserve the lowercase symbol δ for pairwise differences between two such entities (for example, $\delta P = P_i - P_j$ between two amino acids i and j).

We emphasize that P, N, T and Δ are not measured molecular masses. They are exact integer quantities derived from atomic composition, and all arithmetic identities reported in this study are exact integer relations rather than approximate numerical coincidences.

The resulting nucleon values for all twenty canonical amino acids are provided in the dataset `amino_acids_nucleons.csv`, with one row per amino acid and columns for P, N, T and Δ. From this table we additionally precompute three companion datasets of pairwise absolute differences — `amino_acids_proton_differences.csv`, `amino_acids_neutron_differences.csv` and `amino_acids_nucleon_differences.csv` — which group all $\binom{20}{2} = 190$ unordered amino acid pairs by their values of $|\delta P|$, $|\delta N|$ and $|\delta T|$ respectively, where for any pair of amino acids (i, j) we write $\delta P = P_i - P_j$ and analogously for N and T. These companion datasets are used in later sections to identify amino acid pairs whose nucleon differences match specific target values, and in particular to locate pairs that are unique within the twenty amino acids with respect to a given difference.

All nucleon computations performed on codon groups in the remainder of the paper reduce, by linearity, to sums of the per-amino-acid values tabulated in `amino_acids_nucleons.csv`, weighted by codon multiplicities and by the parametrization chosen for the service codons (see Section 2.4).

2.2 Genetic code structure

The standard genetic code establishes a surjective mapping from the 64 codons of the four-letter nucleotide alphabet $\{\mathrm{A}, \mathrm{T}, \mathrm{G}, \mathrm{C}\}$ onto the twenty canonical amino acids and a translation-termination signal. Throughout this paper we use DNA notation (with thymine T rather than uracil U) and adopt the codon assignments of The Standard Code (NCBI translation table 1). Sixty of these codons encode amino acids without additional function; the three codons TAA, TAG and TGA act as stop signals, and the codon ATG, in addition to encoding methionine, serves as the canonical translation initiation signal. Together, these four codons — one start and three stops — are referred to as *service codons* in the remainder of the paper. The full codon-to-product mapping is provided in the dataset `genetic_code_codons.csv`.

A fundamental structural property of the standard code, first identified by Rumer [3, 4], is the partition of the 64 codons into two halves of 32 codons each, distinguished by the degeneracy pattern of the third codon position. The first half, which we denote *Octet I*, consists of the eight codon families in which all four third-position nucleotides encode the same amino acid; these fully four-fold degenerate families are GCN (Ala), CGN (Arg), GGN (Gly), CCN (Pro), ACN (Thr), GTN (Val), CTN (Leu) and TCN (Ser), where N denotes any of the four nucleotides. The second half, denoted *Octet II*, consists of the remaining eight codon families, in which the third-position nucleotide affects the encoded product. Every service codon of the standard code belongs to Octet II, and every service codon carries a purine (A or G) at the third position.

The Rumer partition is not an analytical construct introduced in the present work but a structural feature of the code itself, reflecting both a qualitative asymmetry between fully degenerate and partially degenerate codon families and the quantitative fact that these two qualitatively distinct classes happen to have equal cardinality, eight families and thirty-two codons each. Because Octet I contains no service codons, all nucleon quantities computed over Octet I are independent of any parametrization of start and stop signals and depend only on the nucleon content of the encoded amino acids. The same property holds, for an independent reason, for any codon group whose third-position nucleotide is restricted to pyrimidines (cytosine or thymine), since all four service codons of the standard code carry a purine at the third position. These two conditions — full four-fold degeneracy and purine-free third position — are structurally independent, but both yield the same consequence: nucleon quantities over the affected groups are fixed by the amino acid composition alone and provide parametrization-independent reference values for the arithmetic identities of Section 3.

A further structural property of the Rumer partition, introduced by Rumer [4], is that the two octets are related by a simple permutation of the nucleotide alphabet. Under the transformation $R = (\mathrm{T} \leftrightarrow \mathrm{G},\ \mathrm{C} \leftrightarrow \mathrm{A})$, each codon family of Octet I is mapped to a codon family of Octet II, and vice versa. This transformation, known as the Rumer transformation, rearranges the partition but does not preserve the amino acids encoded by individual codons.

Taken together, the four-fold degeneracy pattern and the Rumer transformation identify the Octet I / Octet II partition as a natural structural division of the standard code, independent of any arithmetic considerations. In the present work we adopt this partition as one of the reference axes along which nucleon quantities are computed and compared.

The codon table (Figure 1) is displayed with its axes in the base ordering C, G, T, A, which is also the default ordering of the accompanying visualizer. This ordering has two independent justifications. Chemically, it was introduced by Rumer [4] on the basis of base type and hydrogen-bond strength: pyrimidines and purines alternate (C, T vs G, A), and within each pair the stronger member precedes the weaker (C before T, G before A). Geometrically, as shown by Panov and Filatov [2], it is one of exactly four orderings (out of 24 possible permutations of the four bases) in which Octet I and Octet II form connected regions on the codon matrix.

To assess the extent to which the arithmetic properties studied in this paper are specific to the standard genetic code, we additionally retrieved the full set of alternative translation tables maintained by NCBI (translation tables numbered 1–33, 27 tables in total, as some numbers are left unused), covering nuclear, mitochondrial and plastid codes across a broad taxonomic range. The codon assignments of all these variants are collected in the dataset `ncbi_genetic_code_registry.csv` and are used in Section 3 for a comparative analysis of the standard code against its known natural alternatives.

2.3 Codon group partitioning

In addition to the Rumer partition introduced in Section 2.2, we employ a second, complementary partitioning scheme based on the chemical identity of the nucleotide at the third codon position. Each of the four nucleotides can be classified along three independent binary chemical axes:

- base type: *purine* $\{\mathrm{A}, \mathrm{G}\}$ versus *pyrimidine* $\{\mathrm{C}, \mathrm{T}\}$;
- hydrogen-bond strength: *strong* (three bonds) $\{\mathrm{C}, \mathrm{G}\}$ versus *weak* (two bonds) $\{\mathrm{A}, \mathrm{T}\}$;
- functional group: *amino* $\{\mathrm{A}, \mathrm{C}\}$ versus *keto* $\{\mathrm{G}, \mathrm{T}\}$.

Each axis splits the four nucleotides into two pairs, giving a total of six two-element nucleotide subsets. Each of these six subsets has a standard one-letter IUPAC ambiguity code (R, Y, S, W, M, K), reflecting that they are the accepted biochemical classifications of the four nucleotides into pairs. For each such subset $X \subset \{\mathrm{A}, \mathrm{T}, \mathrm{G}, \mathrm{C}\}$ with $|X| = 2$, we define the corresponding codon group as the set of all codons whose third-position nucleotide belongs to X.

The six two-nucleotide groups and the four one-nucleotide groups are overlapping rather than mutually disjoint: a given codon belongs to several of them at once. Every codon belongs to exactly three two-nucleotide groups (one per chemical axis) and to exactly one one-nucleotide group.

This third-position partitioning can be applied independently at three levels: to all 64 codons, to the 32 codons of Octet I, and to the 32 codons of Octet II. The same definitions produce groups of different sizes at each level: each two-nucleotide group contains 32 codons at the level of the full code and 16 codons within each octet, and each one-nucleotide group contains 16 codons at the level of the full code and 8 codons within each octet. Where convenient we abbreviate

	C					G					T					A				
C	CCC	Pro	115	62	53	CGC	Arg	174	94	80	CTC	Leu	131	72	59	CAC	His	155	82	73
	CCG	Pro	115	62	53	CGG	Arg	174	94	80	CTG	Leu	131	72	59	CAG	Gln	146	78	68
	CCT	Pro	115	62	53	CGT	Arg	174	94	80	CTT	Leu	131	72	59	CAT	His	155	82	73
	CCA	Pro	115	62	53	CGA	Arg	174	94	80	CTA	Leu	131	72	59	CAA	Gln	146	78	68
G	GCC	Ala	89	48	41	GGC	Gly	75	40	35	GTC	Val	117	64	53	GAC	Asp	133	70	63
	GCG	Ala	89	48	41	GGG	Gly	75	40	35	GTG	Val	117	64	53	GAG	Glu	147	78	69
	GCT	Ala	89	48	41	GGT	Gly	75	40	35	GTT	Val	117	64	53	GAT	Asp	133	70	63
	GCA	Ala	89	48	41	GGA	Gly	75	40	35	GTA	Val	117	64	53	GAA	Glu	147	78	69
T	TCC	Ser	105	56	49	TGC	Cys	121	64	57	TTC	Phe	165	88	77	TAC	Tyr	181	96	85
	TCG	Ser	105	56	49	TGG	Trp	204	108	96	TTG	Leu	131	72	59	TAG	STOP	0	0	0
	TCT	Ser	105	56	49	TGT	Cys	121	64	57	TTT	Phe	165	88	77	TAT	Tyr	181	96	85
	TCA	Ser	105	56	49	TGA	STOP	0	0	0	TTA	Leu	131	72	59	TAA	STOP	0	0	0
A	ACC	Thr	119	64	55	AGC	Ser	105	56	49	ATC	Ile	131	72	59	AAC	Asn	132	70	62
	ACG	Thr	119	64	55	AGG	Arg	174	94	80	ATG	Met	149	80	69	AAG	Lys	146	80	66
	ACT	Thr	119	64	55	AGT	Ser	105	56	49	ATT	Ile	131	72	59	AAT	Asn	132	70	62
	ACA	Thr	119	64	55	AGA	Arg	174	94	80	ATA	Ile	131	72	59	AAA	Lys	146	80	66

Figure 1: Standard genetic code with nucleon parameters of the encoded amino acids, displayed under the Key 0 parametrization (stop codons contribute zero; ATG is treated as methionine). In each cell, the codon appears on the left, the symbol of the encoded product (the three-letter amino acid symbol, or STOP for the three stop codons) at the top center, the total nucleon count T below the product symbol, the proton count P at the top right, and the neutron count N at the bottom right. Codons of Octet I (the eight four-fold degenerate families) are shaded purple; codons of Octet II are shaded yellow. Generated by the accompanying `visualization.html`.

a codon group by its size, writing L_{16} for any group of 16 codons and L_8 for any group of 8 codons, irrespective of the partition that produces it; thus $L_{16} \cap$ Octet I denotes a 16-codon group contained in Octet I, and similarly for the remaining size-and-octet combinations.

The 33 codon groups used in the present work therefore consist of: the full code as a single group of 64 codons; the two octets as groups of 32 codons each; the six two-nucleotide groups and four one-nucleotide groups at the level of the full code; and their analogues within each octet, obtained by intersecting the corresponding chemical group with Octet I or Octet II. We denote such intersections in the natural way: for example, Keto $\cap$ Octet I is the group of 16 codons belonging both to the Keto group and to Octet I, and $\{A\} \cap$ Octet II is the group of 8 codons of Octet II with adenine at the third position. All 33 groups, together with their sizes, codon compositions, and Rumer-octet membership, are listed in the dataset `codon_groups.csv`.

A direct consequence of the code's structure is the following *twin-group property*. Because every codon family in Octet I is fully four-fold degenerate (Section 2.2) and because pyrimidine synonymy holds in all eight Octet II families as well — that is, for every codon family in the standard code, the codons NNC and NNT (with identical first two positions and cytosine versus thymine at the third) encode the same amino acid — the Keto group $\{G, T\}$ and the Strong group $\{C, G\}$ encode the identical multiset of amino acids, and so do the Amino group $\{A, C\}$ and the Weak group $\{A, T\}$. Since all four nucleon quantities P, N, T and Δ depend only on the multiset of encoded products, this yields the identities

$$P(\text{Keto}) = P(\text{Strong}), \qquad N(\text{Keto}) = N(\text{Strong}),$$

$$T(\text{Keto}) = T(\text{Strong}), \qquad \Delta(\text{Keto}) = \Delta(\text{Strong}),$$

and the analogous identities between the Amino and Weak groups. These equalities hold under any parametrization of the service codons, and the same property propagates to the corresponding groups of 16 and 8 codons within each octet. We refer to the Keto/Strong and Amino/Weak pairs as *twin groups*; in the dataset `codon_groups.csv` the twin relationship is recorded in the column `Twin_Group_ID`.

2.4 Parametrization of service codons

The nucleon quantities P, N, T and Δ of any codon group are obtained by summing the corresponding per-amino-acid values from Section 2.1 over all codons in the group. For the 60 sense codons (defined below), this summation is unambiguous: each such codon has a well-defined translation product with a well-defined molecular composition. The remaining four codons — the three stop signals TAA, TAG, TGA and the start signal ATG — require a separate convention, because stop codons do not correspond to any translation product and because the start codon plays a dual role as an initiation signal and as a methionine-encoding codon. We refer to these four codons collectively as *service codons*. The remaining 60 codons we call *sense codons*.[1] The start codon ATG encodes methionine and is a fully functional sense codon at any internal position; it is grouped with the service codons, and excluded from the sense pool, solely in its role as the universal initiation signal, not because it lacks a translation product. Only the ATG initiator is removed on these grounds; every other codon that encodes an amino acid is counted as a sense codon. The sense pool is thus defined as the 64 codons less the three stop codons and the ATG initiator, and its membership does not depend on the choice of parametric key.

To formalize this, we introduce the notion of a *parametric key*: a fixed assignment of nucleon values (P, N) to each of the four service codons. A parametric key specifies how service codons contribute to group sums; it affects only the four service codons, while the nucleon values of the 60 sense codons defined above are fixed by the amino acid composition of Section 2.1 and are

[1] We note that the term *sense codon* is sometimes used in the literature for all 61 non-stop codons, including the ATG initiator. Throughout this paper it denotes the 60 codons remaining after the three stop codons and the ATG initiator are set aside; this usage is fixed in the accompanying datasets and visualizer.

unaffected by the choice of key. In the present study we employ two parametric keys, both of which are recorded in the dataset `key_parameters.csv` under the identifiers `Key 0` and `Key 1`.

Zero parametrization (Key 0). Under the zero parametrization, each of the three stop codons contributes $(P, N) = (0, 0)$, reflecting the fact that a stop codon is not translated into any amino acid and therefore has no molecular product to which nucleon values could be assigned. The start codon ATG is treated as an ordinary methionine codon, with the nucleon values of the free methionine molecule, $(P, N) = (80, 69)$. This convention isolates the arithmetic contribution of the amino-acid-coding portion of the code (with the start codon treated simply as methionine) and is the natural choice when one wishes to study the nucleon structure of the translation without introducing any ad hoc values for non-translated signals.

Symmetrization (Key 1). Under the symmetrization, each of the three stop codons is assigned $(P, N) = (37, 37)$ and the start codon ATG is assigned $(P, N) = (1, 0)$, treated as an initiation marker distinct from methionine. Unlike the zero parametrization, the symmetrization is not adopted by convention but derived analytically as the minimal nonnegative integer solution of a system of symmetry constraints imposed on the four nucleon quantities across the codon groups of Section 2.3. The full derivation, together with the system of constraints and its uniqueness properties, is presented in Section 3; in the present section the symmetrization is introduced purely as a named parametric key to be used in the datasets and the subsequent analysis.

A remark on terminology is in order. The methodological choice underlying the present work is to restrict attention to parameters whose definition is unambiguous and whose interpretation does not depend on any convention beyond the choice of service codon parametrization. The total nucleon count T, the proton count P, and the neutron count N of each amino acid correspond to the actual nuclear composition of the free molecule, restricted to the most abundant stable isotopes (^{1}H, ^{12}C, ^{14}N, ^{16}O, ^{32}S). Once service codons are assigned non-zero contributions to P and N under a parametrization such as Key 1, sums formed over codon groups that include service codons no longer correspond to the nuclear composition of any physical ensemble of molecules. The values continue to be called proton, neutron, and total nucleon counts for consistency with the contributions of the canonical amino acids, but in such sums they are formal arithmetic quantities, and the question of their biochemical meaning does not arise.

The dataset `key_parameters.csv` additionally contains eight reference parametric keys (`Key 2`–`Key 9`), which correspond to various alternative conventions considered during the analysis. They are included for completeness and do not play the role of Key 1, which is derived rather than adopted; they are not used in any of the identities reported in this paper. In the main text we refer to the two principal keys by the descriptive names *zero parametrization* and *symmetrization* respectively, with the dataset identifiers given in parentheses where precision is required.

2.5 Analytical criteria and classification of equalities

The present study systematically examines the nucleon quantities P, N, T and Δ across all codon groups defined in Section 2.3, under each of the two parametric keys of Section 2.4, and records the set of arithmetic properties that these quantities exhibit. To keep the analysis transparent, we organize the search around a fixed hierarchy of criteria stated below.

Criterion 1: Divisibility by 37. For each nucleon quantity P, N, T, Δ of each codon group, we record whether the value is exactly divisible by 37. Divisibility by 37 is the primary criterion of the present work, continuing the line of investigation opened by Shcherbak and Makukov [5] and developed by Panov and Filatov [2]. Where divisibility holds, we additionally record whether it holds in a stricter form — that is, whether the quotient is itself divisible by further small factors. Divisibility by 37 corresponds to a residue of zero modulo 37; in some observations discussed in Section 3, small nonzero residues across structurally related groups are themselves structurally informative. Divisibility expressions for all four nucleon quantities of all codon groups are provided in the datasets `key0_divisibility-37.csv` and `key1_divisibility-37.csv`.

Criterion 2: Roundness in the decimal system. In parallel with Criterion 1, we record whether nucleon quantities are round numbers in the decimal system, that is, exact multiples of a power of ten. This criterion is motivated by the empirical observation that several nucleon quantities of the standard code are multiples of 100 and 1000 — in particular, the parametrization-independent totals over Octet I. Although roundness is stated in decimal notation, it reflects divisibility by the prime factors 2 and 5, which exist independently of the choice of notation; the systematic analysis of divisibility by these and the other primes, together with its relation to the modulus 37, is given in Section 4.2. Where roundness holds, we additionally record whether it holds in a stricter form, as a multiple of a higher power of ten.

Roundness in the decimal system and divisibility by 37 are arithmetically independent conditions, since $\gcd(10, 37) = 1$.

Criterion 3: Equalities between codon groups. For each pair of nucleon quantities taken over two codon groups G_1 and G_2, we record whether the two quantities coincide numerically. Every such equality is classified according to the following three-tier scheme, which depends on the structural type of the equality:

- *Different-size equalities* — the equality relates nucleon quantities of two groups of *different* sizes (for example, a size-64 group and a size-32 group), regardless of whether the two quantities involved are of the same type or of different types. This is the strongest type of equality, because it reflects a coincidence between structurally independent sums computed over codon sets of different scale.

- *Cross-parameter equalities* — the equality relates two *different* nucleon quantities (for example, P of one group and N of another) belonging to groups of the *same* size. This is the intermediate type: the coinciding sums range over codon sets of equal cardinality, but the quantities being compared are structurally distinct.

- *Same-parameter equalities* — the equality relates the *same* nucleon quantity of two distinct groups of the same size (for example, $P(G_1) = P(G_2)$, where G_1 and G_2 have equal numbers of codons). This is the weakest type, since the two sums are of the same structural nature — the same quantity over codon sets of the same cardinality — and coincidence between them is the least informative form of equality. The twin-group identities of Section 2.3 are a special case of this type; they are recorded but treated as structural consequences of Section 2.3 rather than as emergent coincidences.

The classification depends only on the sizes of the compared groups and on the identity of the compared quantities, and not on the specific groups involved. All observed equalities for both parametric keys are tabulated in the datasets `key0_equalities.csv` and `key1_equalities.csv`, in which the three classes are encoded in the `Match_Type` column with values `DIFF_PARAM`, `CROSS_PARAM`, and `SAME_PARAM` respectively. Under Key 0 only same-parameter equalities occur; the stronger different-size and cross-parameter equalities arise only under Key 1.

Criterion 4: Additional structural observations. Beyond divisibility and equalities, we record several further types of structural regularity that do not fit into Criteria 1–3 but nevertheless carry information about the arithmetic organization of the code. These include:

- exact powers — in particular, powers of two or other small integers — realized by nucleon quantities of codon groups;

- anomalously simple reduced fractions arising as ratios P/T, N/T or P/N of nucleon quantities within a single codon group, tabulated in `key0_ratios.csv` and `key1_ratios.csv`;
- coincidences at the level of individual amino acid pairs, in which a nucleon difference $|\delta P|$, $|\delta N|$ or $|\delta T|$ between two amino acids matches a value that appears separately in the group-level analysis (for example, the Trp–Ile pair, for which $|\delta N| = 37$ coincides exactly with the divisor of Criterion 1).

The observations accumulated under Criteria 1–4 are presented in Section 3. They are grouped into parametrization-independent results, which hold for every parametric key, and parametrization-specific results, which depend on the choice of key.

2.6 Computed datasets and reproducibility

The core results reported in the present study are produced by a single reproduction script, `reproduce.py`, that consumes the five input datasets introduced in the previous sections and writes a fixed set of computed output datasets. Two further scripts, the control script `controls.py` and the Monte Carlo script `mc_significance.py` (with its independent re-check `verify_mc.py`), are described separately below and produce the supplementary datasets. This section summarizes the organization of the computed datasets and the reproducibility of the pipeline.

Input datasets. The five input datasets, all of which have been introduced in their respective sections, are: `amino_acids_nucleons.csv` (Section 2.1), `genetic_code_codons.csv` (Section 2.2), `ncbi_genetic_code_registry.csv` (Section 2.2), `codon_groups.csv` (Section 2.3), and `key_parameters.csv` (Section 2.4). These five files fully specify the nucleon composition of the amino acids, the codon assignments of the standard and alternative genetic codes, the partitioning of the 64 codons into groups, and the parametric keys used to handle service codons.

Per-key computed datasets. For each of the two principal parametric keys (Key 0 and Key 1), the pipeline produces four computed datasets with a uniform naming convention, where the prefix `key0_` or `key1_` indicates the active parametric key:

- `*_nucleon-data.csv` — the four fundamental nucleon quantities P, N, T, Δ computed for each of the 33 codon groups defined in `codon_groups.csv` under the active key;
- `*_divisibility-37.csv` — a compact expression of each of the above values as $q \cdot 37 + r$, making divisibility by 37 and residue modulo 37 immediately visible for every group;
- `*_equalities.csv` — the complete enumeration of all numerical equalities between nucleon quantities across codon groups, classified by the `Match_Type` scheme of Section 2.5 into `DIFF_PARAM`, `CROSS_PARAM` and `SAME_PARAM`;
- `*_ratios.csv` — pairwise ratios of the four nucleon quantities for each codon group, provided in both exact fractional form (reduced to lowest terms) and decimal approximation, used for locating anomalously simple fractions as described in Section 2.5.

Together, these eight per-key files contain every numerical value on which the results of Section 3 are based.

Key-independent computed datasets. Three further computed datasets are produced once and do not depend on the choice of parametric key: `amino_acids_proton_differences.csv`, `amino_acids_neutron_differences.csv` and `amino_acids_nucleon_differences.csv`. These enumerate, for all 190 pairs of the twenty canonical amino acids, the absolute differences $|\delta P|$, $|\delta N|$ and $|\delta T|$ respectively, and are used to locate coincidences at the level of individual amino acid pairs as described in Section 2.5.

Sense-pool arithmetic across NCBI codes. The computed dataset `deficit_models_analysis.csv` extends the analysis beyond the standard code by applying the same nucleon bookkeeping to every translation table listed in `ncbi_genetic_code_registry.csv`. The standard code itself appears as one row of the dataset, serving as the reference point against which the remaining 26 tables are compared.

For each translation table the dataset records the global nucleon pools $N(\text{All})$ and $P(\text{All})$ and the sense pools $N(\text{All sense})$ and $P(\text{All sense})$ (total counts over the 60 sense codons of Section 2.4, that is, excluding the three stop codons and the ATG initiator). Two arithmetic tests are then tabulated, each with its own reference structure.

The first test records the deficit of the sense pool neutron pool relative to three reference anchors derived from Octet I. The total nucleon count $T(\text{Octet I})$ serving as each anchor is evaluated under two definitions: a *structural* definition (the eight codon families occupying the Octet I positions in the standard code, regardless of their amino acid assignments in the alternative code) and a *functional* definition (the codon families that are fully four-fold degenerate in the alternative code itself). The three anchors are the constant 3700, the structural $T(\text{Octet I})$ and the functional $T(\text{Octet I})$. For the standard code all three definitions coincide and the deficit equals $111 = 3 \cdot 37$ in all three columns (Section 3.3.2).

The second test records the residue of the sense pool proton pool $P(\text{All sense})$ modulo 37, without reference to any Octet I-based anchor. For the standard code this residue is $-1 \equiv 36 \pmod{37}$, the parity-induced unit shift discussed in Section 3.3.3.

The alternative codes are evaluated against the same two criteria, and the results are used in Section 3.3.5 to assess the extent to which the arithmetic structure of the standard code is shared with its natural variants.

A secondary model, in which the context-dependent stop codons of the three tables 27, 28, and 31 (Karyorelict, Condylostoma, and Blastocrithidia Nuclear Codes) are treated as sense codons rather than as stops, is tabulated in parallel in the `_Alt` columns of the same dataset.

Keto/Amino balance across NCBI codes. The computed dataset `keto_amino_balance_models.csv` extends the Keto/Amino balance analysis of Section 3.3.1 beyond the standard code by applying the same nucleon bookkeeping to every translation table listed in `ncbi_genetic_code_registry.csv`. The standard code itself appears as one row of the dataset, serving as the reference point against which the remaining 26 tables are compared.

For each translation table, the dataset records the sense pool neutron and proton counts of the Keto and Amino half-codes, $N(\text{Keto sense})$, $N(\text{Amino sense})$, $P(\text{Keto sense})$, $P(\text{Amino sense})$, together with their pairwise differences $N(\text{Keto sense}) - N(\text{Amino sense})$ and $P(\text{Keto sense}) - P(\text{Amino sense})$. For the standard code these differences are 37 and 36 respectively (Section 3.3.1); the alternative codes are evaluated against the same pair of reference values, and the results are used in Section 4.1 to identify codes that reproduce the Keto/Amino signature of the standard code.

The same secondary sense-pool model as in the preceding paragraph — in which the context-dependent stop codons of Tables 27, 28, and 31 are returned to the sense pool — is tabulated in parallel in the `_Alt` columns. For the standard code, which contains no context-dependent stop codons, the primary and alternative columns coincide.

Reproducibility. The complete pipeline — input datasets, Python scripts, and all computed output datasets — is provided as a single repository accompanying this paper. Running the script (standard-library Python only, no external dependencies) on the five input datasets regenerates every computed dataset bit-for-bit, and every numerical value appearing in Section 3 can be traced directly to a specific cell of one of these datasets. Within the core pipeline, no randomization is used at any stage of the computation, and all arithmetic underlying the reported regularities is exact, carried out on integer-valued tables and, for ratios, on exact reduced fractions (statistical significance is assessed separately by a fixed-seed Monte Carlo script). The only floating-point values anywhere in the datasets are the rounded decimal approximations printed alongside these exact fractions in the `*_ratios.csv` files, for readability; no result depends on them.

Supplementary visualization tool. An interactive HTML visualization of the genetic code table and its codon-group structure, used during the exploratory phase of this study, is included in the accompanying repository as a supplementary tool. The visualization is not part of the reproducible pipeline described above; any numerical value displayed in its interface under Key 0 or Key 1 that appears in the main text can be verified against the corresponding cell of the computed datasets. An interactive version of the visualizer is available online at https://ruslankhafizov.github.io/genetic-code-arithmetic/.

Supplementary control datasets. Three further datasets probe the specificity of the arithmetic structure and are produced by a separate script `controls.py`, which consumes the same input datasets but is not part of the core pipeline described above. Like that pipeline it uses standard-library Python only, exact integer arithmetic, and no randomization. The dataset `position_axis_analysis.csv` records the four nucleon quantities of six groups — the two sides of each of the three chemical axes — applied independently to the first, second, and third codon position, under the sense pool, Key 0, and Key 1; it is used in Section 4.2 to show that divisibility modulo 37 is concentrated at the third position. The dataset `prime_divisibility_scan.csv` records, for every prime up to 97 and for three value sets — the parametrization-independent groups, Key 0, and Key 1 — how many of the group quantities it divides, over all instances, over distinct values, and over distinct odd cores (the odd part of a number after removing factors of 2). It is used in the same section to show that 37 carries the largest excess apart from 2 and 5. The excess at 2 reflects proton parity, and that at 5 is a trace of the round-number regularity of Criterion 2 (divisibility by 10, 100, and 1000). The excess at 37 rests on more than one independent odd core, whereas the smaller excesses at $73 = 2 \cdot 37 - 1$ and at 61 each reduce to a single odd core and are satellites of an individual quantity. This separation persists even in the value set in which divisibility by 37 is not imposed. The dataset `position_asymmetry_ncbi.csv` records, for each of the 27 NCBI translation tables, the Keto/Amino sense-pool asymmetry for the four nucleon quantities at each codon position, with residues modulo 37; it is used in Section 4.2 to show that this asymmetry is divisible by 37 at all three positions only for the standard code and its exact codon-to-amino-acid equivalent.

Data and code availability. The complete package supporting this work — the LaTeX source of the present preprint, the five input datasets, the thirteen computed datasets, the reproduction script `reproduce.py`, the supplementary control script `controls.py` with its three control datasets, the Monte Carlo script `mc_significance.py` with its summary table and independent re-check `verify_mc.py`, and the interactive visualizer — is available in the accompanying repository at https://github.com/RuslanKhafizov/genetic-code-arithmetic and is archived at Zenodo: DOI: 10.5281/zenodo.20360037.

3 Results

In this section we report the arithmetic properties of the standard genetic code identified in the present study. The standard code is the principal object of analysis throughout; alternative NCBI translation tables are introduced later for comparison and to assess the specificity of the results.

Section 3.1 reports results that do not depend on the parametric keys. Section 3.2 presents results specific to the zero parametrization (Key 0). Section 3.3 derives the symmetrization (Key 1) analytically from a system of symmetry constraints. Section 3.4 reports the arithmetic identities that hold under Key 1, and Section 3.5 summarizes the independent observations across the preceding subsections.

3.1 Parametrization-independent results

Two categories of codon groups in the standard code contain no service codons: Octet I with all its sub-groups, and groups whose third-position nucleotide is restricted to pyrimidines. In addition to these two categories, this subsection covers parametrization-independent equalities between twin pairs of groups, which have a different character: they hold under any key even when the groups participating in the pair themselves contain service codons. More broadly, once the four service codons are excluded, every group becomes parametrization-independent over its remaining sense codons; Figure 2 shows the resulting sums for the sense pool as a whole, across all three levels of partitioning (the graphical layout of the grid is identical across Figures 2, 3, and 4, and its conventions are detailed in the caption of Figure 3).

3.1.1 Arithmetic of Octet I

The four nucleon quantities of Octet I (id 12) are

$$T(\text{Octet I}) = 3700, \qquad P(\text{Octet I}) = 2000,$$

$$N(\text{Octet I}) = 1700, \qquad \Delta(\text{Octet I}) = 300.$$

Notably, all four values are exact multiples of 100, with $P(\text{Octet I})$ additionally being an exact multiple of 1000. Beyond roundness, the total nucleon count $T(\text{Octet I}) = 3700 = 100 \cdot 37$ is also a multiple of 37.

Because every codon family in Octet I is fully four-fold degenerate, the four values above propagate identically to all sub-groups of Octet I: each of the six groups of 16 codons within Octet I (ids 13–18) carries

$$T = 1850, \qquad P = 1000, \qquad N = 850, \qquad \Delta = 150,$$

and each of the four groups of 8 codons within Octet I (ids 19–22) carries

$$T = 925, \qquad P = 500, \qquad N = 425, \qquad \Delta = 75.$$

Divisibility by 37 of T propagates to these levels as well: $1850 = 50 \cdot 37$ and $925 = 25 \cdot 37$, so $T(\text{Octet I})$ is a multiple of 37 at every level of the Octet I hierarchy.

Beyond divisibility of T, the proton and neutron quantities at the 16-codon level of Octet I satisfy $P \equiv +1$ and $N \equiv -1 \pmod{37}$, a systematic ± 1 deviation compensating to zero in $T = P + N$. At the 8-codon level, $\Delta = 75 = 2 \cdot 37 + 1$ shows the same $+1$ deviation directly. This pattern recurs in the pyrimidine-restricted groups of Section 3.1.3 and in parametrization-specific observations of Section 3.2.

The ratios P/T, N/T and Δ/T of Octet I reduce to anomalously simple fractions sharing a common denominator of 37:

$$\frac{P}{T} = \frac{20}{37}, \qquad \frac{N}{T} = \frac{17}{37}, \qquad \frac{\Delta}{T} = \frac{3}{37},$$

and, since every sub-group of Octet I consists of the same four-fold degenerate families in equal proportion, these ratios hold identically at every level of the hierarchy.

Figure 2: Structure of the sense pool. Each group is summed over its non-service codons only; the three stop codons and the initiator ATG are excluded, so every quantity shown is parametrization-independent. (ATG is set aside only in its role as the initiation signal, not because it lacks a translation product.) For each group the four quantities are the total nucleon count T, the proton sum P, the neutron sum N, and the hydrogen count $\Delta = P - N$; green highlights mark divisibility by 37. A single deviation pattern is visible at three nested levels. At the level of the full code the hydrogen count is $\Delta = 296 = 8 \cdot 37$ on the Amino, Weak and Pyrimidine axes but $\Delta = 295$ on Keto, Strong and Purine, one short of a multiple of 37. The same one-unit deviation localizes as the partition refines: among the single-nucleotide groups $\Delta = 148 = 4 \cdot 37$ for $\{A\}$, $\{C\}$ and $\{T\}$ but $\Delta = 147$ for $\{G\}$, the only group containing the ATG position; and among the eight-codon boxes every Octet I box has $\Delta = 75$, while in Octet II three boxes give $\Delta = 73$ and the fourth gives 72 — the box holding the ATG position. The neutron side is aligned with the lattice: N is a multiple of 37 on Keto, Amino, Strong and Weak ($N = 1813 = 49 \cdot 37$ and $N = 1776 = 48 \cdot 37$), and consequently over the whole sense pool $N = 3589 = 97 \cdot 37$. Since N is aligned on all four of these axes, the proton sum $P = N + \Delta$ follows Δ: on Amino and Weak, where Δ lies on the lattice, P does too, whereas on Keto and Strong, where Δ is one unit short, P inherits the same shift, $P \equiv -1 \pmod{37}$.

3.1.2 Pyrimidine synonymy and twin-group identities

In the standard genetic code, the codons NNC and NNT (where C and T are pyrimidine bases) encode the same amino acid.

A direct algebraic consequence of pyrimidine synonymy is the equality of nucleon quantities between the groups **Keto** $\{G, T\}$ (id 2) and **Strong** $\{C, G\}$ (id 3), and between **Amino** $\{A, C\}$ (id 4) and **Weak** $\{A, T\}$ (id 5). Each of these four groups contains 32 codons. The distribution of service codons across them is uneven: Keto and Strong contain the stop codon TAG and the start codon ATG, while Amino and Weak contain two stop codons, TAA and TGA, and no start codon. Each of the four groups thus contains 30 sense codons. Although these groups themselves contain service codons — so their individual nucleon values depend on the parametric key — the service codons enter both groups of each twin pair symmetrically, and the equality of nucleon quantities holds under any parametric key.

Analogous equalities hold between Keto $\cap$ Octet II and Strong $\cap$ Octet II (id 24 and id 25), between Amino $\cap$ Octet II and Weak $\cap$ Octet II (id 26 and id 27), and between the corresponding pairs within Octet I (id 13 and id 14, id 15 and id 16). All these twin pairs are recorded in the `Twin_Group_ID` column of `codon_groups.csv`. Within Octet I, full four-fold degeneracy forces all six 16-codon sub-groups to be mathematically identical, but only the Keto/Strong and Amino/Weak pairs are recorded in `Twin_Group_ID`. The Purine/Pyrimidine pair is excluded because outside Octet I the two groups have substantially different amino acid composition: Purine contains all four service codons, while Pyrimidine contains none. Their equality within Octet I is a degeneracy-specific artifact, not a manifestation of pyrimidine synonymy.

A separate phenomenon of the same origin is the equality of the single-nucleotide groups $\{C\}$ (id 8) and $\{T\}$ (id 10), together with their intersections with each octet. This equality is qualitatively different: it does not relate groups across a chemical axis of partitioning, but follows directly from pyrimidine synonymy itself — a codon with C at the third position and the same codon with T encode the same amino acid by definition. For this reason the single-nucleotide pairs $\{C\}/\{T\}$ are not recorded in `Twin_Group_ID`, which is reserved for pairs whose identical amino acid composition arises from pyrimidine synonymy across a two-nucleotide chemical axis (Keto/Strong or Amino/Weak). $\{C\}$ and $\{T\}$ form the only such pair in the standard code that is entirely free of service codons, and is therefore parametrization-independent both in its values and in their equality.

3.1.3 Arithmetic of pyrimidine-restricted groups

The Pyrimidine group (id 7) and its sub-groups form the second class of parametrization-independent groups of the standard code. Unlike Octet I, these groups span both octets and contain partially degenerate codon families; their arithmetic is therefore not inherited from a single set of values and must be examined separately at each level.

The full pyrimidine group (id 7). The nucleon quantities are

$$T = 4096, \qquad P = 2196, \qquad N = 1900, \qquad \Delta = 296.$$

Three of these four values simultaneously satisfy distinct criteria: $T = 4096 = 2^{12}$ is an exact power of two; $N = 1900$ is an exact multiple of 100; and $\Delta = 296 = 8 \cdot 37$ is divisible by 37.

At the 16-codon level, the groups $\{C\}$ (id 8) and $\{T\}$ (id 10) partition Pyrimidine into two identical halves with values $T = 2048 = 2^{11}$, $P = 1098$, $N = 950$, and $\Delta = 148 = 4 \cdot 37$.

Pyrimidine-restricted groups at the 8-codon level. Intersecting $\{C\}$ and $\{T\}$ with each octet yields four groups of 8 codons whose Δ values, under any parametric key, are

$$\Delta = 75 = 2 \cdot 37 + 1 \quad \text{for } \{C\} \cap \text{Octet I (id 19) and } \{T\} \cap \text{Octet I (id 21)},$$

$$\Delta = 73 = 2 \cdot 37 - 1 \quad \text{for } \{\text{C}\} \cap \text{Octet II (id 30) and } \{\text{T}\} \cap \text{Octet II (id 32)}.$$

These four values exhibit a systematic ± 1 deviation from a multiple of 37: the Octet I values lie one unit above $74 = 2 \cdot 37$, while the Octet II values lie one unit below. Adding the two 8-codon halves {C} and {T} within each octet recovers the intersections of the full pyrimidine group with the corresponding octet:

$$\Delta(\text{Pyrimidine} \cap \text{Octet I}) = 150 = 4 \cdot 37 + 2 \quad \text{(id 18)},$$

$$\Delta(\text{Pyrimidine} \cap \text{Octet II}) = 146 = 4 \cdot 37 - 2 \quad \text{(id 29)}.$$

The ± 1 deviations at the 8-codon level double to ± 2 at the 16-codon level within each octet. Adding across both octets recovers $\Delta(\text{Pyrimidine}) = 296 = 8 \cdot 37$, with the deviations of opposite sign cancelling exactly. The resulting cascade $\pm 1 \to \pm 2 \to 0$ — a systematic small deviation at the 8-codon level that doubles upon intra-octet combination and cancels upon inter-octet combination — recurs in the parametrization-specific observations of Section 3.2.

3.2 Results under the zero parametrization

Under Key 0 — with stop codons contributing zero and ATG treated as methionine — a number of arithmetic properties specific to this parametrization extend the parametrization-independent results of Section 3.1. Figure 3 provides a graphical overview of the quantities computed under Key 0 that are discussed in the subsections below.

3.2.1 Divisibility of T by 37 at the upper levels of partitioning

Under Key 0, the total nucleon count T is divisible by 37 for the four 32-codon groups along the chemical axes Keto/Amino and Strong/Weak:

$$T(\text{Keto}) = T(\text{Strong}) = 4070 = 110 \cdot 37 \quad \text{(id 2 and id 3)};$$

$$T(\text{Amino}) = T(\text{Weak}) = 3848 = 104 \cdot 37 \quad \text{(id 4 and id 5)}.$$

Divisibility of T by 37 extends also to Octet II and its 16-codon sub-groups:

$$T(\text{Octet II}) = 4218 = 114 \cdot 37 \quad \text{(id 23)};$$

$$T(\text{Keto} \cap \text{Octet II}) = T(\text{Strong} \cap \text{Octet II}) = 2220 = 60 \cdot 37 \quad \text{(id 24 and id 25)};$$

$$T(\text{Amino} \cap \text{Octet II}) = T(\text{Weak} \cap \text{Octet II}) = 1998 = 54 \cdot 37 \quad \text{(id 26 and id 27)}.$$

Combined with $T(\text{Octet I}) = 3700 = 100 \cdot 37$ from Section 3.1.1, this gives divisibility by 37 for the total nucleon count of the full code: $T(\text{ALL}) = 7918 = 214 \cdot 37$ (id 1).

The divisibilities $T(\text{Octet I}) = 3700$, $T(\text{Octet II}) = 4218$, and $T(\text{Keto} \cap \text{Octet II}) = T(\text{Strong} \cap \text{Octet II}) = 2220$ by 37 correspond, within our partitioning, to results established by Shcherbak and Makukov [5] and Panov and Filatov [2] (see Section 4.4 for further discussion); the remaining divisibilities of T by 37 given in this subsection follow algebraically from those of these groups. The numerical values of all group quantities under Key 0 are tabulated in `key0_nucleon-data.csv`.

3.2.2 Alignment of Amino and Weak with the lattice of step 37

Under Key 0, the groups Amino (id 4) and Weak (id 5) are the only 32-codon groups of the standard code in which all four nucleon quantities are simultaneously divisible by 37:

$$T = 3848 = 104 \cdot 37, \qquad P = 2072 = 56 \cdot 37,$$

$$N = 1776 = 48 \cdot 37, \qquad \Delta = 296 = 8 \cdot 37.$$

These same groups give anomalously simple reduced ratios:

$$\frac{P}{T} = \frac{7}{13}, \qquad \frac{N}{T} = \frac{6}{13}, \qquad \frac{\Delta}{T} = \frac{1}{13}.$$

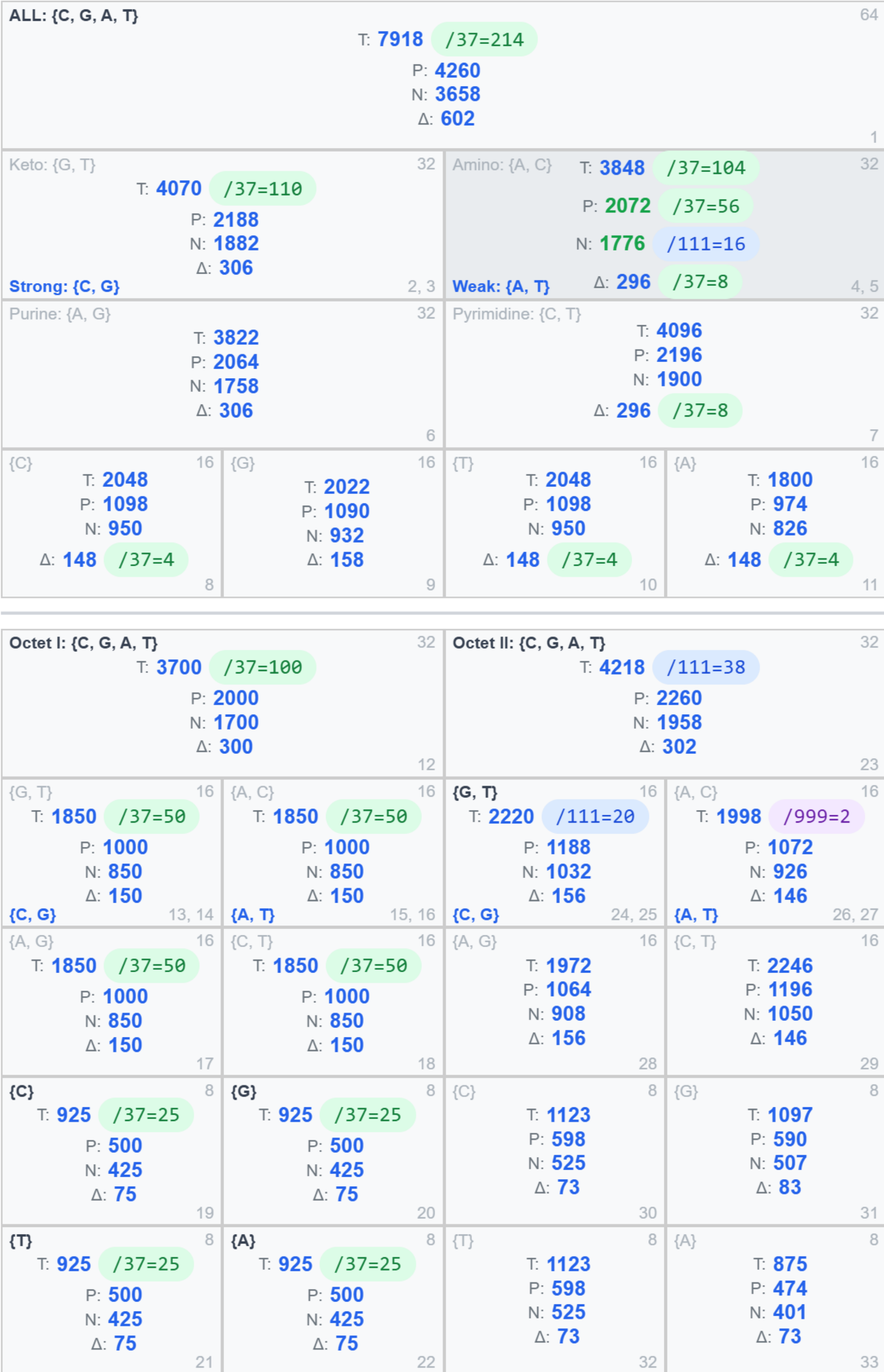

Figure 3: Nucleon parameters (T, P, N, Δ) of all codon groups under Key 0. For each cell: the group name is shown in the upper left; the group size (in codons) in the upper right; the group identifier from `codon_groups.csv` in the lower right. For twin-group pairs (Keto/Strong and Amino/Weak on the global code level, and their analogues within Octet I and Octet II), whose values are identical by pyrimidine synonymy, the display is collapsed into a single block: the second group name appears in the lower left, and the identifiers of both groups are listed, comma-separated, in the lower right. Green highlights indicate exact divisibility by 37; blue, by $111 = 3 \cdot 37$; purple, by $999 = 27 \cdot 37$. Generated by the accompanying `visualization.html`.

3.2.3 The ± 1 deviation pattern under Key 0

The systematic deviation of ± 1 from a multiple of 37, noted in Section 3.1.3 for the pyrimidine-restricted groups, extends under Key 0 to additional groups in parallel positions of the codon hierarchy.

The 16-codon level. The parametrization-independent condition $P \equiv +1$, $N \equiv -1 \pmod{37}$ on the 16-codon sub-groups of Octet I (Section 3.1.1) is complemented, under Key 0, by the mirror condition on the 16-codon sub-groups of the Amino branch of Octet II — $\{\mathrm{A}, \mathrm{C}\} \cap \text{Octet II}$ (id 26) and $\{\mathrm{A}, \mathrm{T}\} \cap \text{Octet II}$ (id 27):

$$P \equiv -1 \pmod{37}, \qquad N \equiv +1 \pmod{37},$$

with $P = 1072 = 29 \cdot 37 - 1$ and $N = 926 = 25 \cdot 37 + 1$.

The 8-codon level. The parametrization-independent part of the pattern at this level is described in Section 3.1.3: $\Delta = +1$ on the 8-codon sub-groups of Octet I, $\Delta = -1$ on the pyrimidine-restricted 8-codon sub-groups of Octet II. Under Key 0, $\{\mathrm{A}\} \cap \text{Octet II}$ (id 33) joins the groups with $\Delta = 73 = 2 \cdot 37 - 1$, completing the mirror triple in Octet II. As a consequence, the parametrization-independent equality $\Delta(\{\mathrm{C}\}) = \Delta(\{\mathrm{T}\}) = 148$ (Section 3.1.3) extends under Key 0 to a third single-nucleotide group at the full 64-codon level: $\Delta(\{\mathrm{A}\}) = 148$ (id 11). The fourth single-nucleotide sub-group of Octet II, $\{\mathrm{G}\} \cap \text{Octet II}$ (id 31), gives $\Delta = 83$ under Key 0 and is the only deviation from the pattern at this level; this group contains both the start codon ATG and the stop codon TAG.

3.3 Analytical derivation of Key 1

The picture presented in Sections 3.1 and 3.2 reveals a systematic feature: under Key 0, the codon groups of the standard code fall into three tiers according to the criteria of Section 2.5. Parametrization-independent groups (Octet I and the pyrimidine-restricted groups) exhibit a high degree of order; groups containing only stop codons (Amino, Weak) align with the lattice of step 37 across all four nucleon quantities; groups containing the start codon ATG deviate from the pattern. The deviations track ATG specifically: within the unaligned Keto and Strong, the ATG-free sub-group is itself aligned in its proton-neutron difference ($\Delta(\{\mathrm{A}\}) = 148 = 4 \cdot 37$, id 11), whereas the sub-groups containing ATG deviate ($\Delta(\{\mathrm{G}\} \cap \text{Octet II}) = 83$, id 31, and $\Delta(\{\mathrm{G}\}) = 158$, id 9). In each case the deviation traces to ATG, treated under Key 0 as methionine with nucleon values $(80, 69)$ that do not lie on the lattice of step 37. The degree of alignment thus depends on which service codons, if any, a given group contains. This points to the possibility of a different parametrization, under which the codon groups exhibit a higher degree of symmetry. In this section we derive such a parametrization analytically from a system of symmetry conditions.

The presentation is divided into five subsections. Section 3.3.1 traces the macroscopic Keto/Amino asymmetry of the standard code to a single pair of canonical amino acids, Trp–Ile, and establishes the arithmetic uniqueness of this pair, including the parity theorem for protons that explains why $|\delta P| = 37$ is impossible for any pair. In Sections 3.3.2 and 3.3.3 the parameters of the stop and start codons are determined sequentially through symmetry conditions — first for neutrons, then for protons. Section 3.3.4 shows that the solution obtained is the minimal non-negative integer element of an infinite family of solutions. Section 3.3.5 tests the parametrization against alternative NCBI translation tables as a comparative check of its specificity for the standard code.

3.3.1 Localization of the Keto/Amino asymmetry in the Trp–Ile pair

At the level of the full code under Key 0, the differences between the nucleon quantities of Keto and Amino are

$$\begin{aligned} T(\text{Keto}) - T(\text{Amino}) &= 4070 - 3848 = 222 = 6 \cdot 37, \\ P(\text{Keto}) - P(\text{Amino}) &= 2188 - 2072 = 116, \\ N(\text{Keto}) - N(\text{Amino}) &= 1882 - 1776 = 106, \\ \Delta(\text{Keto}) - \Delta(\text{Amino}) &= 306 - 296 = 10. \end{aligned}$$

The total nucleon difference is itself a multiple of 37; its decomposition into proton and neutron components is not.

Localization at the family level. Of the 16 codon families, 13 contain no service codon, and in each of these the codons NNG (Keto branch) and NNA (Amino branch) encode the same amino acid; the Keto and Amino branches of these families contribute equally to both sums. One additional family, TA, contributes equally to both branches: its pyrimidine pair TAC/TAT encodes Tyr in both branches, and its purine pair TAG/TAA consists of two stop codons, one in each branch, which cancel from the Keto/Amino difference under any parametrization assigning equal values to the three stop codons. Asymmetry is realized only in two families:

- the family AT, where ATG = Met (Keto) versus ATA = Ile (Amino);
- the family TG, where TGG = Trp (Keto) versus TGA = Stop (Amino).

In the family TG, Cys is present symmetrically in both branches and does not contribute; in the family AT, two of the three Ile codons (ATT and ATC) are likewise distributed symmetrically. The asymmetry of the full code therefore reduces to the nucleon counts of three molecules:

$$T(\text{Keto}) - T(\text{Amino}) = T(\text{Trp}) + T(\text{Met}) - T(\text{Ile}) = 204 + 149 - 131 = 222.$$

Reduction to the Trp–Ile pair. The start codon ATG enters the asymmetry above through its translation product Met under Key 0, and is the only service codon contributing to the difference under that parametrization. Excluding the service codons from the balance, so that the sense pools of Keto and Amino are compared directly, gives:

$$\begin{aligned} T(\text{Keto sense}) - T(\text{Amino sense}) &= 3921 - 3848 = 73, \\ P(\text{Keto sense}) - P(\text{Amino sense}) &= 2108 - 2072 = 36, \\ N(\text{Keto sense}) - N(\text{Amino sense}) &= 1813 - 1776 = 37, \\ \Delta(\text{Keto sense}) - \Delta(\text{Amino sense}) &= 295 - 296 = -1. \end{aligned}$$

Here $X(G \text{ sense})$ denotes the sum of the quantity $X \in \{T, P, N, \Delta\}$ over the sense codons of G, that is, after removing all service codons from G. Since the source of the asymmetry is now a single pair of canonical amino acids, Trp and Ile, the differences of the sense pools are exactly the nucleon differences $T(\text{Trp}) - T(\text{Ile})$, $P(\text{Trp}) - P(\text{Ile})$, $N(\text{Trp}) - N(\text{Ile})$ and $\Delta(\text{Trp}) - \Delta(\text{Ile})$ of this pair.

Arithmetic uniqueness of the Trp–Ile pair. Although the nucleon differences of Trp–Ile are determined solely by the nucleon composition of the two molecules and have no direct relation to the structure of the code, both proton and neutron components of the difference turn out to be coordinated with the number 37, which plays the role of an organizing multiplier in the arithmetic of the standard code: $|\delta N(\text{Trp}, \text{Ile})| = 37$ coincides with this number exactly, $|\delta P(\text{Trp}, \text{Ile})| = 36$ is offset from it by one. The two are related by $\delta P = \delta N - (\Delta_{\text{Ile}} - \Delta_{\text{Trp}}) = 37 - 1 = 36$: the proton difference lies one unit below the neutron difference because isoleucine carries one more hydrogen than tryptophan. Parity forces the proton difference to be even, so δP cannot reach 37. Parity dictates only that it is even; that it is exactly 36 rather than 34 or 38 is a specific chemical property of these two molecules, and it is this exact value that later lets the service parameters resolve to $p_{\text{stop}} = n_{\text{stop}} = 37$ (Section 3.3.4).

The value $|\delta N| = 37$. Among the 190 pairs of canonical amino acids, the value $|\delta N| = 37$ is realized by only two pairs: Trp–Ile and Trp–Leu. The latter pair gives the same value because Leu and Ile are isomers with identical nucleon counts $P = 72$, $N = 59$; the pair structurally realized via the families AT and TG is Trp–Ile. The full distribution of $|\delta N|$ over the 190 pairs is given in `amino_acids_neutron_differences.csv`.

The value $|\delta P| = 36$. Among the 190 pairs of canonical amino acids, the value $|\delta P| = 36$ is realized only by the same two pairs — Trp–Ile and Trp–Leu. The full distribution of $|\delta P|$ over the 190 pairs is given in `amino_acids_proton_differences.csv`. This offset by exactly one unit is not coincidental: proton parity forbids the odd value 37 for any pair, as shown below, leaving 36 and 38 as the nearest realizable values, of which the standard code realizes the lower, 36.

The induced value $|\delta T| = 73$. The total nucleon difference of the pair satisfies $|\delta T| = |\delta P| + |\delta N| = 36 + 37 = 73 = 2 \cdot 37 - 1$ and is therefore not an independent observation. For completeness, $|\delta T| = 73$ is realized by exactly the same two pairs, Trp–Ile and Trp–Leu, among the 190 pairs of canonical amino acids; the full distribution of $|\delta T|$ is given in `amino_acids_nucleon_differences.csv`.

Parity theorem for protons. Every one of the 20 canonical amino acids has an even proton count. Direct inspection of the table of nucleon quantities confirms this: each value of P in `amino_acids_nucleons.csv` is even. Consequently, the difference $|\delta P|$ is always even for any pair of amino acids, and in particular $|\delta P| = 37$ is impossible for any pair. The nearest even values to 37 are 36 and 38, and the pair realized in the standard code, Trp–Ile, gives $|\delta P| = 36$: a deficit of one relative to 37, rather than an excess. This parity is a consequence of the electronic structure of the canonical amino acids. In their neutral ground states, they are closed-shell molecules and therefore contain an even number of electrons. Since a neutral molecule has equal numbers of electrons and protons, the total proton count is even as well.

3.3.2 Determination of neutron parameters

We seek a parametrization of the service codons under which the residual deviations described in Section 3.3 disappear. We denote the (yet unknown) parameters of the stop codons by $(p_{\text{stop}}, n_{\text{stop}})$ and of the start codon by $(p_{\text{start}}, n_{\text{start}})$, with the convention that the three stop codons share the same parameters — a hypothesis already invoked in Section 3.3.1 when reducing the Keto/Amino asymmetry to the Trp–Ile pair. In this subsection we determine the neutron components n_{stop} and n_{start}. The proton components are determined in Section 3.3.3.

Distribution of service codons. All four service codons belong to Octet II; Octet I contains no service codon. Along the Keto/Amino axis, the start codon ATG and the stop codon TAG carry third-position G and belong to the Keto branch, while the stop codons TAA and TGA carry third-position A and belong to the Amino branch. Thus the Keto branch contains one stop and one start, and the Amino branch contains two stops and no start. The same distribution applies to the Strong and Weak branches respectively, by the Keto $\equiv$ Strong and Amino $\equiv$ Weak identities of Section 3.1.2.

Balance equation for Keto and Amino. The neutron pools of Keto and Amino, expressed through the sense pool values established in Section 3.3.1 and the unknown parameters of the service codons, are

$$N(\text{Keto}) = 1813 + n_{\text{start}} + n_{\text{stop}},$$
$$N(\text{Amino}) = 1776 + 2\,n_{\text{stop}}.$$

Imposing the symmetry condition $N(\text{Keto}) = N(\text{Amino})$ (the Purine/Pyrimidine axis cannot serve here, as its balance is incompatible with divisibility by 37; see Section 3.3.4) yields

$$n_{\text{stop}} - n_{\text{start}} = 37.$$

The right-hand side is the difference $N(\text{Keto sense}) - N(\text{Amino sense})$ established in Section 3.3.1; its value is determined by the nucleon difference of the Trp–Ile pair. The condition is a one-parameter family of solutions in two unknowns; a second independent constraint is required to fix the parameters individually.

Divisibility of the global neutron pool by 37. The sense pool of all 60 sense codons of the standard code carries

$$N(\text{All sense}) = N(\text{Octet I}) + N(\text{Octet II sense}) = 1700 + 1889 = 3589 = 97 \cdot 37.$$

This exact divisibility by 37 is an inherent chemical property of the sense codons and is independent of any service-codon parametrization. However, at the level of the entire genetic code, the inclusion of service codons disrupts this arithmetic alignment. Under Key 0, where the start codon is interpreted as Met, the global pool becomes $N(\text{All}) = 3658 = 98 \cdot 37 + 32$, which is not divisible by 37. We require the analytical parametrization to restore this underlying divisibility at the global scale:

$$N(\text{All}) = 3589 + n_{\text{start}} + 3\,n_{\text{stop}} \equiv 0 \pmod{37},$$

which reduces to

$$n_{\text{start}} + 3\,n_{\text{stop}} \equiv 0 \pmod{37}.$$

Solution. Substituting $n_{\text{stop}} = n_{\text{start}} + 37$ from the first condition into the second yields

$$4\,n_{\text{start}} + 111 \equiv 0 \pmod{37}.$$

Since $111 = 3 \cdot 37 \equiv 0 \pmod{37}$ and $\gcd(4, 37) = 1$, this reduces to

$$n_{\text{start}} \equiv 0 \pmod{37},$$

giving the family of integer solutions $n_{\text{start}} = 0, 37, 74, \ldots$ The minimal non-negative integer solution is

$$n_{\text{start}} = 0, \quad n_{\text{stop}} = 37.$$

The structure of the solution family and the minimality of this choice are addressed in Section 3.3.4.

Emergent cross-parameter equality. Substitution of the obtained values into Octet II yields

$$N(\text{Octet II}) = 1889 + 0 + 3 \cdot 37 = 2000 = P(\text{Octet I}).$$

This cross-parameter equality between the neutron content of Octet II and the proton content of Octet I was not imposed during the derivation; it emerges as a consequence of the minimal non-negative integer solution to the two conditions imposed (Keto/Amino balance and global divisibility by 37).

Non-triviality of the integer solution. Three arithmetic facts combine in the derivation, and each is independent of the other two:

1. the difference $N(\text{Keto sense}) - N(\text{Amino sense}) = 37$, a property of the Trp–Ile pair (Section 3.3.1);
2. the divisibility of the global sense pool $N(\text{All sense}) = 3589 = 97 \cdot 37$ by 37, a chemical property of the standard code independent of any service-codon parametrization;
3. the existence of a minimal non-negative integer solution with both parameters small. The relative primality $\gcd(4, 37) = 1$, where 4 is the coefficient of n_{start} in the combined equation and 37 is the modulus, ensures that the divisibility condition reduces to $n_{\text{start}} \equiv 0 \pmod{37}$ rather than to a constraint admitting multiple small residues.

Macroscopic consequences. Under the values $n_{\text{stop}} = 37$, $n_{\text{start}} = 0$, the neutron pools of the four 32-codon groups along the chemical axes satisfy

$$N(\text{Keto}) = N(\text{Amino}) = N(\text{Strong}) = N(\text{Weak}) = 1850 = \tfrac{1}{2}\, T(\text{Octet I}),$$

and the global identity

$$N(\text{All}) = T(\text{Octet I}) = 3700$$

holds. The neutron content of each chemical-axis half of the code coincides with exactly half of the T-invariant of Octet I, and the global neutron pool coincides with this T-invariant in full. The complete set of consequences is collected in Section 3.4.

3.3.3 Determination of proton parameters

Having determined the neutron components in Section 3.3.2, we now apply the same approach to determine the proton components p_{stop} and p_{start}. The convention that the three stop codons share the same parameters is preserved.

Balance equation for Keto and Amino. The proton pools of Keto and Amino, expressed through the sense pool values established in Section 3.3.1 and the unknown parameters of the service codons, are

$$P(\text{Keto}) = 2108 + p_{\text{start}} + p_{\text{stop}},$$
$$P(\text{Amino}) = 2072 + 2\, p_{\text{stop}}.$$

Imposing the corresponding condition $P(\text{Keto}) = P(\text{Amino})$ (again on the Keto/Amino axis, for the reason noted above) yields

$$p_{\text{stop}} - p_{\text{start}} = 36.$$

The right-hand side is the difference $P(\text{Keto sense}) - P(\text{Amino sense})$ established in Section 3.3.1; its value is 36 rather than 37, the parity-constrained value of $|\delta P(\text{Trp}, \text{Ile})|$ (Section 3.3.1). The condition is a one-parameter family of solutions in two unknowns; a second independent constraint is required to fix the parameters individually.

Divisibility of the global proton pool by 37. The sense pool of all 60 sense codons of the standard code carries

$$P(\text{All sense}) = P(\text{Octet I}) + P(\text{Octet II sense}) = 2000 + 2180 = 4180 = 113 \cdot 37 - 1.$$

Unlike the neutron pool, the global proton pool of the sense codons is not divisible by 37; it lies one proton below the nearest multiple, $P(\text{All sense}) \equiv -1 \pmod{37}$. We require the analytical parametrization of the service codons to close this deficit and bring the global proton pool onto the lattice of step 37:

$$P(\mathrm{All}) = 4180 + p_{\mathrm{start}} + 3\,p_{\mathrm{stop}} \equiv 0 \pmod{37},$$

which reduces to

$$p_{\mathrm{start}} + 3\,p_{\mathrm{stop}} \equiv 1 \pmod{37}.$$

Solution. Substituting $p_{\mathrm{stop}} = p_{\mathrm{start}} + 36$ from the first condition into the second yields

$$4\,p_{\mathrm{start}} + 108 \equiv 1 \pmod{37}.$$

Since $108 = 2 \cdot 37 + 34$ and $\gcd(4, 37) = 1$, this reduces to

$$4\,p_{\mathrm{start}} \equiv 4 \pmod{37}, \quad \text{hence} \quad p_{\mathrm{start}} \equiv 1 \pmod{37},$$

giving the family of integer solutions $p_{\mathrm{start}} = 1, 38, 75, \ldots$ The minimal non-negative integer solution is

$$p_{\mathrm{start}} = 1, \quad p_{\mathrm{stop}} = 37.$$

The structure of the solution family and the minimality of this choice are addressed in Section 3.3.4.

Emergent equality of stop parameters. The neutron and proton parameters of the stop codons obtained independently in Sections 3.3.2 and 3.3.3 coincide:

$$p_{\mathrm{stop}} = n_{\mathrm{stop}} = 37.$$

Each stop codon under Key 1 carries the same number of protons and neutrons, both equal to the organizing multiplier 37. This equality was not imposed; it follows from the minimal non-negative integer solutions to two formally independent systems — one in neutrons, one in protons — each constrained by Keto/Amino balance and global divisibility by 37.

Non-triviality of the integer solution. Three arithmetic facts combine in the derivation:

1. the difference $P(\text{Keto sense}) - P(\text{Amino sense}) = 36$, the parity-constrained value of $|\delta P(\mathrm{Trp}, \mathrm{Ile})|$ (Section 3.3.1);
2. the residue $P(\text{All sense}) \equiv -1 \pmod{37}$, a deficit of one proton on the 37-lattice and a chemical property of the standard code independent of any service-codon parametrization;
3. the existence of a minimal non-negative integer solution with both parameters small. The relative primality $\gcd(4, 37) = 1$, where 4 is the coefficient of p_{start} in the combined equation and 37 is the modulus, ensures that the divisibility condition reduces to $p_{\mathrm{start}} \equiv 1 \pmod{37}$ rather than to a constraint admitting multiple small residues.

Facts (1) and (2) share a common origin in the proton parity of the canonical amino acids: parity excludes odd values on the 37-lattice — both the value 37 for $|\delta P|$ and an odd multiple for $P(\text{All sense})$ — forcing an unavoidable deviation from these target values. The particular values that this deviation takes (36 and -1) are fixed by the nucleon composition of the code. The minimal solution $p_{\mathrm{start}} = 1$ closes exactly this single-proton deficit at the global scale.

Macroscopic consequences. Under the values $p_{\text{stop}} = 37$, $p_{\text{start}} = 1$, the proton pools of the four 32-codon groups along the chemical axes satisfy

$$P(\text{Keto}) = P(\text{Amino}) = P(\text{Strong}) = P(\text{Weak}) = 2146 = \tfrac{1}{2}\,T(\text{Octet II}),$$

and the global identity

$$P(\text{All}) = T(\text{Octet II}) = 4292$$

holds, with $T(\text{Octet II})$ evaluated under Key 1. Unlike $T(\text{Octet I}) = 3700$, which is parametrization-independent and serves as a chemical invariant, $T(\text{Octet II})$ depends on the values assigned to the service codons; under Key 1 it takes the value 4292. Together with the analogous neutron identities $N(\text{All}) = T(\text{Octet I}) = 3700$ and $N(\text{Keto}) = \frac{1}{2}\,T(\text{Octet I})$ established in Section 3.3.2, the parametrization realizes a dual matching: the global neutron pool aligns with $T(\text{Octet I})$, the global proton pool aligns with $T(\text{Octet II})$, and each chemical-axis half of the code coincides with exactly half of the corresponding octet's total nucleon content. The complete set of consequences is collected in Section 3.4.

3.3.4 Hierarchy of constraints and the choice of Key 1

The derivation in Sections 3.3.2 and 3.3.3 imposed two kinds of constraint on the service-codon parameters, each applied to neutrons and to protons: Keto/Amino balance and divisibility of the global pool by 37. Each constraint reduces the dimension of the solution space, and the family of admissible parametrizations narrows in three steps:

1. Keto/Amino balance alone, applied separately to neutrons and protons, gives a two-parameter family of solutions: the conditions $n_{\text{stop}} - n_{\text{start}} = 37$ and $p_{\text{stop}} - p_{\text{start}} = 36$ reduce the four service-codon parameters to two independent degrees of freedom, one for each nucleon quantity. All elements of this family align the four 32-codon groups along the chemical axes Keto / Amino / Strong / Weak.

2. Adding global divisibility of $N(\text{All})$ by 37 further restricts the neutron parameters to the discrete one-parameter family $n_{\text{start}} \in \{0, 37, 74, \ldots\}$, that is $n_{\text{start}} = 37k$ for $k \in \mathbb{Z}_{\geq 0}$. Adding global divisibility of $P(\text{All})$ by 37 similarly restricts the proton parameters to $p_{\text{start}} \in \{1, 38, 75, \ldots\}$, that is $p_{\text{start}} = 1 + 37m$ for $m \in \mathbb{Z}_{\geq 0}$. The combined family becomes a discrete two-dimensional lattice indexed by $(k, m) \in \mathbb{Z}^2_{\geq 0}$ with step 37 in each direction.

3. Selecting the minimal non-negative integer element of this lattice ($k = m = 0$) gives Key 1: $(p_{\text{stop}}, n_{\text{stop}}) = (37, 37)$, $(p_{\text{start}}, n_{\text{start}}) = (1, 0)$.

The minimal element is distinguished from all other solutions by a combination of structural features that the constraints alone do not require: the value $p_{\text{stop}} = n_{\text{stop}} = 37$, equal to the organizing multiplier of the code and attained only at the minimal element; the value $p_{\text{start}} = 1$, closing exactly the parity-induced deficit $P(\text{All sense}) \equiv -1 \pmod{37}$ (Section 3.3.3); and the value $n_{\text{start}} = 0$, at which the start codon contributes no neutrons. Of these, only $p_{\text{stop}} = n_{\text{stop}} = 37$ singles out the minimal element on its own: $n_{\text{start}} = 0$ holds for every element with $k = 0$ and $p_{\text{start}} = 1$ for every element with $m = 0$, and the two coincide with $p_{\text{stop}} = n_{\text{stop}} = 37$ only at $k = m = 0$. Higher elements of the lattice satisfy the same constraints but lack this combination. We adopt Key 1 as the parametrization of the standard genetic code throughout the remainder of the paper.

The stop-parameter equality as a two-axis condition. The equality $p_{\text{stop}} = n_{\text{stop}}$ and the offset $p_{\text{start}} - n_{\text{start}} = 1$, obtained in Section 3.3.4 from the minimal solution of the two modulo-37 systems, also follow from the two chemical-axis balances alone, with no reference to 37. Writing $\Delta = P - N$, the Keto/Amino balance ($n_{\text{stop}} - n_{\text{start}} = 37$, $p_{\text{stop}} - p_{\text{start}} = 36$) gives

$\Delta_{\text{start}} = \Delta_{\text{stop}}+1$. Since all four service codons lie in the Purine branch, equalizing the Δ-invariant across the Purine/Pyrimidine axis requires the service increment to close the sense-pool deficit $\Delta(\text{Pyr sense}) - \Delta(\text{Pur sense}) = 1$, that is $\Delta_{\text{start}} + 3\,\Delta_{\text{stop}} = 1$. The same purine localization of the service codons is also what forces the balance in P and N to be imposed on the Keto/Amino axis rather than on Purine/Pyrimidine: with all increments confined to the Purine branch, a Purine = Pyrimidine balance would require $p_{\text{start}} + 3\,p_{\text{stop}} = 212$ and $n_{\text{start}} + 3\,n_{\text{stop}} = 211$, whereas global divisibility requires these same sums to be $\equiv 1$ and $\equiv 0 \pmod{37}$; since $212 \equiv 27$ and $211 \equiv 26$, the two demands are incompatible, and among the third-position axes only a balance on Keto/Amino (equivalently, by the twin identity of Section 3.1.2, on Strong/Weak) is compatible with the 37-lattice. The obstruction is specific to P and N, whose service sums are large (212 and 211) and land at the wrong residues. For $\Delta = P - N$ the situation differs: the Purine/Pyrimidine Δ-deficit is only $\Delta(\text{Pyr sense}) - \Delta(\text{Pur sense}) = 1$, small enough to be closed by the service increment, and it coincides with the offset $\Delta_{\text{start}} - \Delta_{\text{stop}} = 1$ left by the Keto/Amino balance. The Keto/Amino P/N balance and the Purine/Pyrimidine Δ-balance are therefore mutually compatible, which is what makes the two-axis condition available.

This fixes only the proton–neutron differences of the service codons, not their absolute values: the modulus 37 and the minimality condition remain necessary to obtain $(37, 37)$ and $(1, 0)$. Its content is that the equality of the stop parameters is not tied to the modulus—it holds whenever the Keto/Amino and Purine/Pyrimidine Δ-balances hold—and that it reduces to the agreement of two sense-pool unit offsets: the proton–neutron difference $\delta P - \delta N = -1$ of the localizing pair (Section 3.3.1) and the branch deficit $\Delta(\text{Pyr sense}) - \Delta(\text{Pur sense}) = +1$. The coincidence is thus relocated rather than removed, but it is shown to be a joint property of two chemical axes rather than an artifact of the parametrization.

Convergence of independent routes. The values of Key 1 are not tied to a single derivation. Three routes, resting on different structural inputs, reach the same assignment. The first is the modulo-37 system used above: the Keto/Amino balance together with divisibility of the global neutron and proton pools, whose minimal nonnegative solution is $(p_{\text{stop}}, n_{\text{stop}}) = (37, 37)$, $(p_{\text{start}}, n_{\text{start}}) = (1, 0)$. The second is the two-axis condition of Section 3.3.4: the Keto/Amino balance together with the Δ-balance across the Purine/Pyrimidine axis forces $p_{\text{stop}} = n_{\text{stop}}$ and $p_{\text{start}} = n_{\text{start}}+1$ with no reference to the modulus. The third fixes the neutron parameters directly from the parametrization-independent half-octet target $N(\text{Amino}) = N(\text{Keto}) = \frac{1}{2}\,T(\text{Octet I}) = 1850$: the sense-pool deficits $1850 - 1776 = 74 = 2 \cdot 37$ (closed by the two stop codons of the Amino branch) and $1850 - 1813 = 37$ (closed by the stop and start of the Keto branch) give $n_{\text{stop}} = 37$ and $n_{\text{start}} = 0$.

3.3.5 Comparative test against alternative NCBI translation tables

The standard genetic code is one of 27 translation tables in the NCBI registry. The derivation of Key 1 in Sections 3.3.2 and 3.3.3 relies on two arithmetic properties of the sense pool of the standard code:

$$N(\text{All sense}) = 3589 = 97 \cdot 37, \quad P(\text{All sense}) = 4180 \equiv -1 \pmod{37}.$$

The first is the global divisibility of the neutron pool by 37, which equals $T(\text{Octet I}) - N(\text{All sense}) = 3700 - 3589 = 111 = 3 \cdot 37$, the deficit closed by three stop codons under Key 1 to align $N(\text{All})$ with the parametrization-independent invariant $T(\text{Octet I})$. The second is the parity-induced deficit of one proton on the 37-lattice, which is closed by the start-codon parameter $p_{\text{start}} = 1$. Whether these two properties are specific to the standard code or are shared by its known variants is a question that admits a direct comparative test on each of the two nucleon quantities independently.

Method. For each translation table, two quantities are computed: the neutron deficit Δ_N relative to a reference total T_{ref}, and the proton residue $P(\text{sense pool}) \bmod 37$. Both quantities are computed under two model choices for the sense pool:

- Sense pool, Model 1 (primary): all codons listed as stops in the NCBI registry are excluded, even when they are read as amino acids in context-dependent codes. The primary start codon ATG is excluded as the universal initiator. Initiation is a property of the first codon position rather than of the triplet itself, so exactly one initiator codon is removed; alternative start codons such as GTG and TTG carry their elongation meaning at every internal position and are counted as sense codons.

- Sense pool, Model 2 (alternative): in the three codes with context-dependent stops (Tables 27, 28, 31), codons that function as amino acids in mid-sequence and as stops at gene termini are treated as sense codons. For all other tables, Model 2 coincides with Model 1. A detailed description of the model conventions is given in the dataset documentation accompanying `deficit_models_analysis.csv`.

For the neutron deficit Δ_N, three model choices of T_{ref} are also considered: the constant $T(\text{Octet I}) = 3700$ of the standard code, the total nucleon content of the eight structural Octet I families under the amino-acid assignments of the specific table, and the total nucleon content of all 4-fold degenerate codon families in the specific table. For the proton residue, the relevant quantity is intrinsic to the sense pool itself and does not require a reference total. The Key 1 parametrization closes the deficits and matches the residues for the standard code; a comparative test asks for which other tables the same arithmetic structure holds. The full set of computed values across all tables and model combinations is tabulated in `deficit_models_analysis.csv`.

Results. The two tests give parallel outcomes. In each test, the arithmetic property holds in exactly the same three tables:

- Table 1 (the standard code) and Table 11 (the Bacterial, Archaeal and Plant Plastid Code) have $\Delta_N = 111 = 3 \cdot 37$ under all three reference models, and $P(\text{sense pool}) \equiv -1 \pmod{37}$ under both sense-pool models. Their amino acid assignments are identical, and they share the same set of stop codons; arithmetically they are the same code applied in different domains.

- Table 28 (the Condylostoma Nuclear Code) gives the same values as the standard code under Model 1, and different values under Model 2. The coincidence under Model 1 arises because the three context-dependent stop codons of Condylostoma, once excluded alongside ATG, leave a sense codon set identical to the one used in the standard code analysis. Under Model 2, the three context-dependent codons are returned to the sense pool, and neither the deficit nor the residue matches the standard code.

In the remaining 24 tables, neither property holds under any model combination: Δ_N is not divisible by 37, and $P(\text{sense pool}) \bmod 37$ takes values different from -1. The two tests identify the same set of tables, but this reflects a single underlying sense pool rather than independent agreement between the two conditions: Tables 1, 11, and (under Model 1) 28 all share the standard code's sense-codon assignments.

Interpretation. The Key 1 parametrization is not a generic feature of genetic codes. Across 27 known translation tables and six reasonable definitions of the neutron deficit, plus two definitions of the proton residue, only the standard code and its exact codon-to-amino-acid equivalent (Table 11) satisfy both arithmetic conditions under every model. One additional table (Table 28) matches the standard code under the primary stop-codon interpretation but loses the match under the alternative reading of its context-dependent codons. All other tables fail both conditions under every model: they either reassign codons within the structural Octet I families, shift the set of 4-fold-degenerate families, or use a different number of stop codons, and in each case the arithmetic match between the sense pool and the lattice of step 37 is destroyed. The two arithmetic conditions, taken together, single out the standard genetic code among all known translation tables.

3.4 Consequences under Key 1

In this section we describe the arithmetic structure of the standard genetic code under Key 1: alignment of the 32-codon chemical-axis groups, identities between the global nucleon pools and the structural totals of the octets, higher-order divisibility properties, the Δ-invariant across all six 32-codon groups, and the Purine/Pyrimidine asymmetry. Figure 4 shows the same overview computed under Key 1. The four chemical-axis groups Keto, Strong, Amino, and Weak now become identical in T, P, N, and Δ, accounting for the visible uniformity in the top half of the figure.

3.4.1 Alignment of the chemical-axis 32-codon groups

Under Key 1, the four 32-codon groups along the chemical axes Keto/Amino and Strong/Weak are aligned in all four nucleon quantities to the common values

$$T = 3996, \quad P = 2146, \quad N = 1850, \quad \Delta = 296.$$

The equality Keto = Amino follows from the balance conditions imposed during the derivation of n_{stop} and p_{stop}; the identities Keto $\equiv$ Strong and Amino $\equiv$ Weak are parametrization-independent (Section 3.1.2). The numerical values of all group quantities under Key 1 are tabulated in `key1_nucleon-data.csv`.

Of the three axes that partition the codon table by third position — Keto/Amino, Strong/Weak, and Purine/Pyrimidine — only the first two yield this alignment. The Purine/Pyrimidine axis behaves differently and is analyzed in Section 3.4.5.

Rational signature. The alignment can be restated in projective form. Under Key 1, the whole code and the four chemical-axis 32-codon groups share a single rational profile of nucleon ratios,

$$\frac{P}{T} = \frac{29}{54}, \qquad \frac{N}{T} = \frac{25}{54}, \qquad \frac{\Delta}{T} = \frac{2}{27},$$

with $P : N = 29 : 25$. The two terms of this ratio are $T(L_{16} \cap \text{Octet II})/(2 \cdot 37) = 29$ and $T(L_{16} \cap \text{Octet I})/(2 \cdot 37) = 25$, so the profile carries the same pairing of P with Octet II and N with Octet I as the macroscopic identities of Section 3.4.2. The ratio profiles of all groups are tabulated in `key1_ratios.csv`.

3.4.2 Macroscopic identities and propagation to chemical-axis halves

Under Key 1, the global nucleon pools of the standard code satisfy three identities relating them to the structural totals of the two octets:

$$\begin{aligned} N(\text{All}) &= T(\text{Octet I}) = 3700, \\ P(\text{All}) &= T(\text{Octet II}) = 4292, \\ N(\text{Octet II}) &= P(\text{Octet I}) = 2000. \end{aligned}$$

The first identity records the cross-parameter equality already noted in Section 3.3.2: the global neutron pool of the code under Key 1 equals the parametrization-independent chemical constant $T(\text{Octet I})$. The second is the analogue on the proton side, with $P(\text{All}) = 4292$ matching

Figure 4: The same overview as Figure 3, computed under the analytically derived parametrization Key 1, in which all four service codons are treated as formal parameters: the three stop codons contribute $(P, N) = (37, 37)$ each, and ATG contributes $(P, N) = (1, 0)$ (the methionine value of ATG is not used). Under this parametrization the four chemical-axis 32-codon groups Keto, Strong, Amino, Weak become identical in T, P, N, and Δ (top half of the figure). Generated by the accompanying `visualization.html`.

T(Octet II) evaluated under Key 1; unlike T(Octet I), the right-hand side here depends on the service-codon parametrization. The third identity restates the first at the level of individual octets: since $T(\text{Octet I}) = P(\text{Octet I}) + N(\text{Octet I})$ tautologically, with chemical constants $P(\text{Octet I}) = 2000$ and $N(\text{Octet I}) = 1700$, the equality $N(\text{All}) = T(\text{Octet I})$ from the first identity gives $N(\text{Octet II}) = 2000 = P(\text{Octet I})$.

Combining the macroscopic identities with the alignment of the four chemical-axis groups (Section 3.4.1) gives

$$N(\text{Keto}) = N(\text{Amino}) = N(\text{Strong}) = N(\text{Weak}) = 1850 = \tfrac{1}{2}\,T(\text{Octet I}),$$
$$P(\text{Keto}) = P(\text{Amino}) = P(\text{Strong}) = P(\text{Weak}) = 2146 = \tfrac{1}{2}\,T(\text{Octet II}).$$

Each chemical-axis half of the code carries exactly half of the structural total of the corresponding octet. The same two values reappear at the L_{16} level inside the octets. Each of the six L_{16} sub-groups of Octet I carries $T = 1850$, equal across the six sub-groups because each 4-fold degenerate family of Octet I contributes the same amino acid nucleon to every L_{16} partition. Inside Octet II, the four L_{16} sub-groups along the chemical axes Keto/Amino and Strong/Weak carry $T = 2146$ under Key 1; the two L_{16} sub-groups along the Purine/Pyrimidine axis are not aligned with this value (Section 3.4.5). The cross-parameter equality $N(\text{Octet II}) = P(\text{Octet I}) = 2000$ similarly resolves at the L_{16} level: each L_{16} sub-group of Octet I carries $P = 1000$, and each chemical-axis L_{16} sub-group of Octet II carries $N = 1000$ under Key 1. The full set of equalities arising under Key 1, classified by relation type, is tabulated in `key1_equalities.csv`.

A parametrization-independent counterpart on the proton side. The value 2000 that appears in the cross-parameter identity $N(\text{Octet II}) = P(\text{Octet I}) = 2000$ also arises elsewhere in the arithmetic of the code, on the proton side and without reference to any service-codon parametrization. The sense pool of Octet II carries $P(\text{Octet II sense}) = 2180$; subtracting the single pair Trp and Ile that localizes the Keto/Amino asymmetry (Section 3.3.1) gives

$$P(\text{Octet II sense}) - \big(P(\text{Trp}) + P(\text{Ile})\big) = 2180 - 180 = 2000 = P(\text{Octet I}).$$

The proton excess of Octet II over Octet I at the sense-pool level coincides exactly with the proton count of the localizing pair: $P(\text{Trp}) + P(\text{Ile}) = 180 = P(\text{Octet II sense}) - P(\text{Octet I})$. The analogous identity for neutrons does not hold — $N(\text{Trp}) + N(\text{Ile}) = 155$, while $N(\text{Octet II sense}) - N(\text{Octet I}) = 189$, a residual difference of 34 — and the proton equality is specific to the proton component of the chemistry of the canonical amino acids.

3.4.3 Divisibility by 37 and by 999

The arithmetic of Key 1 is organized by two divisibility properties of the standard code. The divisibility by 37 was imposed during the derivation of n_{stop} and p_{stop} for the global pools N(All) and P(All); under Key 1 it extends to many further group quantities. The divisibility by 999 is not required by the system and holds independently for the total nucleon pool.

Divisibility by 37. Under Key 1, divisibility by 37 holds for the global neutron and proton pools, and for the 32-codon groups along the chemical axes Keto/Amino and Strong/Weak in T, P, N, and Δ:

$$\begin{aligned}
N(\text{All}) &= 100 \cdot 37,\\
P(\text{All}) &= 116 \cdot 37,\\
T(\text{Keto}) = T(\text{Amino}) = T(\text{Strong}) = T(\text{Weak}) &= 108 \cdot 37,\\
P(\text{Keto}) = P(\text{Amino}) = P(\text{Strong}) = P(\text{Weak}) &= 58 \cdot 37,\\
N(\text{Keto}) = N(\text{Amino}) = N(\text{Strong}) = N(\text{Weak}) &= 50 \cdot 37,\\
\Delta(\text{Keto}) = \Delta(\text{Amino}) = \Delta(\text{Strong}) = \Delta(\text{Weak}) &= 8 \cdot 37.
\end{aligned}$$

By the first two macroscopic identities of Section 3.4.2, divisibility by 37 also extends to the global proton-neutron imbalance: $\Delta(\text{All}) = P(\text{All}) - N(\text{All}) = T(\text{Octet II}) - T(\text{Octet I}) = 16 \cdot 37 = 592$. The full set of divisibility records under Key 1 is tabulated in `key1_divisibility-37.csv`.

Divisibility by 999. Under Key 1, the global total nucleon pool of the standard code and each of the four chemical-axis 32-codon groups in T satisfy

$$T(\text{All}) = 7992 = 8 \cdot 999, \quad T(\text{Keto}) = T(\text{Amino}) = T(\text{Strong}) = T(\text{Weak}) = 3996 = 4 \cdot 999.$$

The system of conditions imposed in Section 3.3 requires divisibility by 37 but not by the higher modulus $999 = 27 \cdot 37$. In factored form $T(\text{All}) = 2^3 \cdot 3^3 \cdot 37 = 6^3 \cdot 37$. The divisibility of $T(\text{All})$ and $T(\text{half})$ by 999 is an independent property of the standard code under Key 1.

3.4.4 The Δ-invariant across all six 32-codon groups

Under Key 1, all six 32-codon groups defined by the third codon position carry the same value of the proton-neutron difference:

$$\Delta(\text{Keto}) = \Delta(\text{Amino}) = \Delta(\text{Strong}) = \Delta(\text{Weak}) = \Delta(\text{Purine}) = \Delta(\text{Pyrimidine}) = 296.$$

For the four groups along the chemical axes Keto/Amino and Strong/Weak, this follows from the alignment in P and N established in Section 3.4.1: equal P and equal N imply equal Δ. For the Purine and Pyrimidine groups, which are not aligned in T, P, or N (Section 3.4.5), the Δ-equality is independent of the alignment: the P-shift and the N-shift between the two groups coincide in magnitude, so that their difference $\Delta = P - N$ is preserved across all three axes.

The common value $\Delta = 296$ resolves through a cascade joining the lower partition levels. At the 8-codon level, every $L_8 \cap \text{Octet I}$ carries $\Delta = 75$ as a parametrization-independent property of the 4-fold-degenerate families (Section 3.1.1), and every $L_8 \cap \text{Octet II}$ carries $\Delta = 73$ under Key 1 (paralleling the Δ-pattern of Section 3.1.3):

$$\Delta(L_8 \cap \text{Octet I}) = 75, \qquad \Delta(L_8 \cap \text{Octet II}) = 73.$$

The 16-codon level reproduces these values along two distinct partitions of the code. Within each octet, every L_{16} sub-group inherits twice the corresponding L_8 value,

$$\Delta(L_{16} \cap \text{Octet I}) = 150, \qquad \Delta(L_{16} \cap \text{Octet II}) = 146.$$

Across octets, each single-base 16-codon group of the full code joins one $L_8 \cap \text{Octet I}$ with one $L_8 \cap \text{Octet II}$, giving

$$\Delta(\{\text{C}\}) = \Delta(\{\text{G}\}) = \Delta(\{\text{T}\}) = \Delta(\{\text{A}\}) = 75 + 73 = 148 = 4 \cdot 37$$

at All-code level (ids 8–11). This four-way equality is specific to Key 1: under Key 0 the placement of the start codon ATG as methionine in {G} shifts $\Delta(\{\text{G}\})$ to 158, breaking the symmetry with the remaining three single-base groups. The two 16-codon paths converge at the 32-codon level: the common value $\Delta = 296 = 2 \cdot 148 = 150 + 146$ obtains either as the sum of two single-base 16-codon groups along any axis, or equivalently as the sum of an $L_{16} \cap \text{Octet I}$ and an $L_{16} \cap \text{Octet II}$ from the same axis.

3.4.5 Purine/Pyrimidine asymmetry

The Purine/Pyrimidine axis is the one chemical axis on which the standard code is not aligned in T, P, or N. Under Key 1,

$$\begin{aligned} N(\text{Pyrimidine}) - N(\text{Purine}) &= 100, \\ P(\text{Pyrimidine}) - P(\text{Purine}) &= 100, \\ T(\text{Pyrimidine}) - T(\text{Purine}) &= 200, \\ \Delta(\text{Pyrimidine}) - \Delta(\text{Purine}) &= 0. \end{aligned}$$

The asymmetry is localized in Octet II. Within Octet I, Purine and Pyrimidine contain equal proton and neutron pools as a parametrization-independent consequence of 4-fold degeneracy (Section 3.1); the imbalance arises entirely from Octet II, where all four service codons reside in the Purine branch (third position A or G).

The shifts in P and N are equal in magnitude (100 in each), and this magnitude equality is what preserves the Δ-invariant across the Purine and Pyrimidine groups (Section 3.4.4). At the level of the sense pool alone, the two shifts are not equal:

$$P(\text{Pyr sense}) - P(\text{Pur sense}) = 212, \quad N(\text{Pyr sense}) - N(\text{Pur sense}) = 211,$$

with a residual proton-neutron asymmetry of one in favor of P. Under Key 1, the service codons add 112 protons and 111 neutrons to the Purine branch: one ATG with $p_{\text{start}} = 1$, $n_{\text{start}} = 0$, plus three stop codons each with $p_{\text{stop}} = n_{\text{stop}} = 37$. The proton-neutron asymmetry of the service increments matches that of the sense pool. Adding these increments to the Purine branch reduces the Pyr–Pur proton difference from 212 to 100 and the neutron difference from 211 to 100, equalizing the two macroscopic shifts.

Purine synonymy at the single-base level under Key 1. At the 16-codon level, the four single-base groups of the full code split into two pairs by their (T, P, N) profiles:

$$T(\{\text{C}\}) = T(\{\text{T}\}) = 2048, \quad P(\{\text{C}\}) = P(\{\text{T}\}) = 1098, \quad N(\{\text{C}\}) = N(\{\text{T}\}) = 950,$$
$$T(\{\text{G}\}) = T(\{\text{A}\}) = 1948, \quad P(\{\text{G}\}) = P(\{\text{A}\}) = 1048, \quad N(\{\text{G}\}) = N(\{\text{A}\}) = 900,$$

while Δ remains uniform across all four groups (Section 3.4.4). The pyrimidine equality $\{\text{C}\} \equiv \{\text{T}\}$ is parametrization-independent and follows from the universal C/T synonymy at the family level (Section 3.1.2). The purine equality $\{\text{G}\} \equiv \{\text{A}\}$ is specific to Key 1, with the mechanism local to Octet II: the branches $\{\text{G}\} \cap$ Octet II and $\{\text{A}\} \cap$ Octet II contain asymmetrically distributed service codons (ATG and TAG in $\{\text{G}\}$ versus TAA and TGA in $\{\text{A}\}$), and the symmetrized Key 1 values cancel the corresponding sense-pool asymmetries exactly:

$$\{\text{G}\} \cap \text{Octet II} = \{\text{A}\} \cap \text{Octet II} = (T = 1023,\ P = 548,\ N = 475).$$

The Pyrimidine pair and the Purine pair differ by exactly 100 in T and 50 in each of P and N; doubling these shifts reproduces the macroscopic Purine/Pyrimidine asymmetry at the 32-codon level.

Inside Octet II at the 16-codon level. The L_{16} sub-groups along the Purine/Pyrimidine axis carry

$$N(\text{Pur} \cap \text{Octet II}) = 950, \quad N(\text{Pyr} \cap \text{Octet II}) = 1050,$$
$$P(\text{Pur} \cap \text{Octet II}) = 1096, \quad P(\text{Pyr} \cap \text{Octet II}) = 1196,$$

showing a ± 50 deviation from the values carried by the L_{16} sub-groups of Octet II along the Keto/Amino and Strong/Weak axes, which under Key 1 are $N = 1000$ and $P = 1146$ (Section 3.4.2). The Purine and Pyrimidine branches of Octet II thus straddle the values of the other two axes symmetrically.

3.5 Synthesis

The arithmetic structure described in Sections 3.1–3.4 separates into two levels: properties that hold independently of any parametrization of the service codons, and properties that appear under the parametrization Key 1. Within the second level, a further distinction is necessary between relations that follow algebraically from the conditions imposed in the derivation of Key 1 and relations that hold under Key 1 without following from those conditions. This subsection collects the results under these headings, with particular attention to the arithmetic preconditions that made the derivation possible in the first place.

Preconditions of the derivation. The derivation of Key 1 rests on a structural framework in which the codon table is partitioned into Octet I and Octet II by the four-fold degeneracy of codon families, and further partitioned along the chemical axes Keto/Amino, Strong/Weak, and Purine/Pyrimidine, as well as into single-base groups, by the third codon position. Within this framework, several properties of the standard code combine to make the parametric system of Sections 3.3.2–3.3.3 solvable in small non-negative integers; these properties are themselves the most consequential findings of the work.

After exclusion of the service codons, the entire Keto/Amino asymmetry of the code localizes to a single pair of canonical amino acids, Trp–Ile, whose nucleon differences $|\delta N| = 37$ and $|\delta P| = 36$ are realized among the 190 pairs of canonical amino acids only by this pair and by Trp–Leu, the latter giving the same values because Leu and Ile are nucleon-equivalent isomers; the value $|\delta N| = 37$ matches the organizing multiplier exactly, and $|\delta P| = 36$ falls one unit short, consistent with the even-proton property of canonical amino acids that forbids $|\delta P| = 37$ for any pair (Section 3.3.1). The sense pool of the code carries $N(\text{All sense}) = 97 \cdot 37$ and $P(\text{All sense}) = 113 \cdot 37 - 1$, the same parity-induced unit shift that appears in $|\delta P(\text{Trp}, \text{Ile})|$; the neutron divisibility is specific to the sense pool and is absent under Key 0 (Section 3.3.2). The four service codons of the standard code distribute across the chemical axes as one stop and one start in Keto, two stops in Amino, producing the balance conditions; in the global pool the same four codons—one start and three stops—make the divisibility condition a residue equation whose coefficient at the start parameter is 4. The divisibility deficits then close arithmetically: the neutron deficit $T(\text{Octet I}) - N(\text{All sense}) = 3 \cdot 37 = 111$ is a multiple both of the number of stops and of the modulus 37, the proton residue -1 is the smallest possible non-zero contribution from the start codon, and the relative primality $\gcd(4, 37) = 1$ ensures that the combined system has a unique minimal solution rather than multiple small residues. The neutron and proton systems are formally independent, yet their minimal solutions coincide at the value $p_{\text{stop}} = n_{\text{stop}} = 37$. The comparative test against the 26 alternative NCBI translation tables (Section 3.3.5) shows that the arithmetic match between the deficit and the lattice of step 37 holds only for the standard code and its exact codon-to-amino-acid equivalent.

Algebraic consequences of the Key 1 system. The balance conditions imposed during the derivation directly equate the Keto and Amino groups in N and P. Combined with the parametrization-independent identities Keto $\equiv$ Strong and Amino $\equiv$ Weak (Section 3.1.2), this aligns the four 32-codon groups along the Keto/Amino and Strong/Weak axes in all four nucleon quantities (T, P, N, Δ) (Section 3.4.1). From the alignment follow the half-code identities $N(\text{half}) = \frac{1}{2}T(\text{Octet I})$ and $P(\text{half}) = \frac{1}{2}T(\text{Octet II})$, and the divisibility of their nucleon quantities by 37.

Independent observations under Key 1. The remaining identities hold under the parametrization without following from the balance conditions. The global neutron pool reaches $N(\text{All}) = 3700$, the minimal admissible multiple of 37, which coincides with the parametrization-independent constant $T(\text{Octet I})$; the equivalent statement at the level of individual octets is $N(\text{Octet II}) = P(\text{Octet I}) = 2000$, a cross-parameter equality between a parametric quantity and a chemical constant. The analogous proton identity $P(\text{All}) = T(\text{Octet II}) = 4292$ holds under the same parametrization, completing the dual pairing between the global nucleon pools and the two octets, and follows from the neutron identity together with $T = P + N$ (Section 3.4.2). The total nucleon pool satisfies $T(\text{All}) = 8 \cdot 999$, a divisibility by the higher modulus 999 not present among the imposed conditions (Section 3.4.3). The Δ-invariant $\Delta = 296$ extends to all six 32-codon groups, including the Purine and Pyrimidine groups, which are not aligned in T, P or N (Section 3.4.4). On the Purine/Pyrimidine axis, the proton and neutron shifts between the two groups are equal and both take the value 100, with the equality of the two shifts following from the matching of the parity-induced unit shift between the sense pool and the parametric increments (Section 3.4.5).

At the single-base level, the purine equality $\{G\} \equiv \{A\}$ holds under Key 1 in addition to the parametrization-independent pyrimidine equality $\{C\} \equiv \{T\}$. A parametrization-independent counterpart on the proton side gives $P(\text{Octet II sense}) - \big(P(\text{Trp}) + P(\text{Ile})\big) = P(\text{Octet I}) = 2000$: the proton excess of Octet II over Octet I on the sense pool coincides exactly with the proton mass of the localizing pair, an identity that does not hold for neutrons.

A proton-side offset of one unit. In the parametrization-independent arithmetic, two neutron quantities meet the 37-lattice exactly while the corresponding proton quantities fall one unit short. The sense pool carries $N(\text{All sense}) = 97 \cdot 37$ against $P(\text{All sense}) = 113 \cdot 37 - 1$; the single pair Trp–Ile that localizes the Keto/Amino asymmetry has $|\delta N| = 37$ against $|\delta P| = 36$. The derived parametrization then closes the offset with one extra proton on the start codon, $p_{\text{start}} = 1$, where the neutron side requires none, $n_{\text{start}} = 0$. The common cause is the even-proton property of the canonical amino acids (Section 3.3.1): a sum or difference of even proton counts is even and cannot equal an odd multiple of 37, so both 37 itself and $113 \cdot 37$ are unreachable on the proton side. That the miss is exactly one unit in each case, rather than a larger even amount, is a property of the specific nucleon counts and is not fixed by parity alone.

Two recurrent integers: 37 and the round-number scale. The integer 37 appears at both levels of the structure and in roles not directly connected by the derivation. At the parametrization-independent level it is the common denominator of the Octet I ratios, the divisor of the sense pool $N(\text{All sense}) = 97 \cdot 37$, and the value of $|\delta N(\text{Trp}, \text{Ile})|$. Under Key 0 it organizes the alignment of the Amino and Weak groups and their sub-group $\{A\}$ with the 37-lattice (Section 3.2). Under Key 1 it is the modulus of the imposed divisibility conditions, the common value of the stop-codon parameters, and a divisor of the nucleon quantities of the aligned 32-codon groups. These occurrences meet at the cross-parameter identity $N(\text{All}) = T(\text{Octet I})$, where the parametrization-independent value $100 \cdot 37$ and the Key 1-level divisibility of $N(\text{All})$ by 37 fall on the same number. That a topological quantity of Octet I, a global neutron sum, and the neutron difference of a single amino-acid pair are all expressed through the same modulus, although none of these is derived from the others, indicates that the regularities form a connected whole rather than separate coincidences; the identities relating them, such as $T(\text{Octet I}) - N(\text{All sense}) = 3 \cdot |\delta N(\text{Trp}, \text{Ile})|$, follow once the shared modulus is granted and add no independent divisibility of their own.

A second integer scale is also visible in the arithmetic of the code. The parametrization-independent constants of Octet I — $P(\text{Octet I}) = 2000$, $N(\text{Octet I}) = 1700$, $\Delta(\text{Octet I}) = 300$ — are all multiples of 100, with $P(\text{Octet I})$ a multiple of 1000. Under Key 1 the cross-parameter equality $N(\text{Octet II}) = P(\text{Octet I}) = 2000$ brings the neutron content of Octet II onto the same scale, and the neutron pools of the Purine and Pyrimidine groups satisfy $N(\text{Purine}) = 1800$ and $N(\text{Pyrimidine}) = 1900$. The Purine/Pyrimidine shifts in P and N take the value 100 at the 32-codon level and ± 50 at the L_{16} level. These round-number occurrences do not reduce to the multiplier 37 and constitute a separate arithmetic regularity of the standard code under Key 1.

4 Discussion

This section examines how far the regularities reported above are specific to the standard code and its structure, rather than artifacts of the analysis. Section 4.1 tests the robustness of the Trp–Ile localization across the alternative NCBI codes; Section 4.2 asks whether the third codon position and the modulus 37 are distinguished from other positions and divisors; and Section 4.3 assesses the concentration on the 37-lattice against a random-code null. Section 4.4 situates the results relative to prior work, and Section 4.5 states the limitations and outlook.

4.1 Robustness of the Trp–Ile localization across alternative genetic codes

Section 3.3.5 tested whether the arithmetic structure underlying the derivation of Key 1 is specific to the standard code, and found that the matching of the sense-pool deficit with the 37-lattice holds only for the standard code and its exact codon-to-amino-acid equivalent. A separate question concerns the robustness of one of the inputs to the derivation: the localization of the Keto/Amino asymmetry in the single pair Trp–Ile (Section 3.3.1). The two tests probe different aspects of the arithmetic of the standard code, and they identify different sets of matching alternative codes.

A comparative analysis across the 27 NCBI translation tables, computed in `keto_amino_balance_models.csv` shows that the sense-pool differences $N(\text{Keto sense}) - N(\text{Amino sense}) = 37$ and $P(\text{Keto sense}) - P(\text{Amino sense}) = 36$ are reproduced exactly in seven tables. Five tables — Table 1 (Standard), Table 6 (Ciliate, Dasycladacean and Hexamita Nuclear), Table 11 (Bacterial, Archaeal and Plant Plastid), Table 29 (Mesodinium Nuclear), and Table 30 (Peritrich Nuclear) — reproduce the $(37, 36)$ signature under both sense-pool models. The remaining two, Table 27 (Karyorelict Nuclear) and Table 28 (Condylostoma Nuclear), reproduce it only under the primary interpretation of context-dependent stop codons.

The mechanism is structural and operates differently in different tables. Table 11 has an amino acid assignment identical to the standard code and shares its set of stop codons, so the $(37, 36)$ signature holds trivially. In Tables 6, 29, and 30, the codons TAG and TAA are reassigned to the same amino acid (Gln, Tyr, or Glu respectively); since TAG sits in the Keto branch of family TA and TAA in the Amino branch, this symmetric reassignment contributes identically to both branches of the sense pool and the Keto/Amino differences are preserved. No codons other than TAA and TAG are reassigned in these tables; in particular, neither TGG (Trp) nor any codon of Ile is changed, and the Trp/Ile localization established in Section 3.3.1 continues to hold.

Tables 27 and 28 involve context-dependent stop codons and behave differently under the two sense-pool models. Table 28 declares all three of TAA, TAG, TGA as context-dependent stops; under Model 1, where these codons are excluded as stops, its sense pool is identical to that of the standard code, and the $(37, 36)$ signature is reproduced. Under Model 2, the three codons are returned to the sense pool with their context-dependent amino acid translations, the sense pool composition changes, and the signature is lost. Table 27 permanently reassigns TAA and TAG to Gln (the same symmetric mechanism as Tables 6, 29, 30) and treats only TGA as a context-dependent stop; under Model 1 this preserves the $(37, 36)$ signature, while under Model 2 the inclusion of TGA as Trp in the Keto branch breaks it.

Translation tables that reassign ATA (typically to Met in mitochondrial codes) or TGA (typically from stop to Trp) change the amino acid content of the carrier codons and no longer reproduce the $(37, 36)$ signature.

The asymmetry between nuclear and mitochondrial codes is informative. All seven tables matching the $(37, 36)$ signature are nuclear or bacterial; none of the thirteen mitochondrial translation tables in the NCBI registry reproduces it. In each case the failure traces to the reassignment of ATA or TGA, both reassignments altering codons on which the localization depends.

A finer decomposition of the test connects the seven matching tables to the two comparative tests. In the standard code, the four sense-pool half-quantities have the residues $(N_{\text{Keto}}, N_{\text{Amino}}, P_{\text{Keto}}, P_{\text{Amino}}) \equiv (0, 0, -1, 0) \pmod{37}$, an arithmetic profile that combines divisibility of each half-pool by 37 with the parity-induced unit shift on the proton side. Tables 1, 11, and 28 (under Model 1) reproduce all four residues exactly. For Table 11 this follows from identity with the standard code; for Table 28 under Model 1 it follows from the sense pool being identical to that of the standard code once the context-dependent stops are excluded. These are the same three tables that match in the deficit test of Section 3.3.5.

The four additional tables matching the $(37, 36)$ signature (6, 27, 29, 30) exhibit a different arithmetic profile: N_{Keto} and N_{Amino} share the same non-zero residue mod 37, and P_{Keto} and P_{Amino} differ by one unit, but the absolute level has detached from the 37-lattice. This is the arithmetic signature of symmetric reassignment of TAA and TAG to a common amino acid: each half-pool acquires the same non-zero contribution from the new amino acid, so the Keto/Amino differences are preserved while the divisibility of each half by 37 is lost. Among the 27 NCBI tables, 13 preserve the divisibility of at least one half-pool by 37, three of them mitochondrial (Tables 4, 16, and 23).

The two comparative tests therefore probe distinct aspects of the arithmetic of the standard code. The deficit test of Section 3.3.5 measures the absolute alignment of the global sense pool with the 37-lattice and matches only tables arithmetically equivalent to the standard code in their sense pool (Tables 1, 11, and 28 under Model 1). The localization test of this subsection measures the preservation of a relational pattern between the Keto and Amino branches and additionally matches tables in which the only divergence from the standard code is a symmetric reassignment of TAA and TAG (Tables 6, 27, 29, 30).

4.2 Specificity of the third codon position and of the modulus 37

Two features are built into the framework of this work rather than derived from the data: the three chemical axes are read off the third codon position (Section 2.3), and divisibility by 37 is the primary criterion (Section 2.5). Two deterministic controls test whether the observed structure is specific to these two choices or would arise generically.

Codon position. Applying the same three chemical axes to the first and second codon position, instead of the third, removes the structure almost entirely. At the third position the Amino and Weak groups have all four nucleon quantities divisible by 37, the Keto and Strong groups have their neutron pool divisible, and the Pyrimidine group has Δ divisible. At the first position the only divisibility is the neutron pool of the Keto/Amino axis, present in both its complementary halves, and this is not an independent fact: Keto and Amino are complementary, so their sense-pool neutron counts sum to $N(\text{All sense}) = 3589 = 97 \cdot 37$ (Section 3.3.2), and divisibility of one half forces divisibility of the other. At the second position no nucleon quantity of any axis is divisible by 37. The control thus shows that divisibility by 37 along the chemical axes is confined to the third codon position: the same axes applied to the first and second positions do not meet the 37-lattice, the single first-position exception being the dependent one noted above. This excludes the alternative in which any equal-size split of the sense pool into chemical halves would align with the 37-lattice irrespective of position. The full values for all three positions are tabulated in `position_axis_analysis.csv`.

Choice of modulus. For every prime up to 97 we count how many of the four nucleon quantities of the codon groups it divides, over three value sets: the parametrization-independent groups (those with no service codon, on which 37 cannot be imposed), the groups under Key 0 (which does not impose 37), and the groups under Key 1 (whose derivation does). The scan ranges over primes rather than all integers because divisibility by a composite is the conjunction of divisibilities by its prime-power factors and introduces no new modulus, and because raising a prime to a higher power lowers the chance rate $1/m$ and thereby inflates the excess over the same quantities without adding information; the prime is the appropriate unit. The informative statistic is accordingly the excess over the chance rate $1/p$, computed over distinct values so that quantities forced to coincide by the structure of the code are not counted more than once.

Three primes divide their share at essentially the chance rate (3, 7 and 13). Two exceed it modestly, by less than a factor of three across the three value sets: 2, because every canonical amino acid has an even proton count (Section 3.3.1), and 5, reflecting the round-number regularity already reported in Section 3.5, which the scan thus corroborates. Against this background 37 divides five of the twenty-eight distinct parametrization-independent quantities, thirteen of the seventy distinct Key 0 quantities, and ten of the forty-seven under Key 1 — in each case several times the naive rate 1/37. This excess ratio is descriptive, not a p-value: the scan is deterministic and has no null model, and the nested group hierarchy makes the quantities non-independent, so the ratio alone cannot be read as a significance. The load of the argument rests instead on a structural property the ratio does not capture, developed below: that the excess for 37 rests on more than one independent odd core, whereas the comparable factors 73 and 61 each reduce to a single core. Because the effect is already present in the value sets where 37 is not imposed, it is in any case not an artifact of the Key 1 derivation.

Three features set 37 apart from the two small primes, all independent of the number base. It is a larger prime, so each divisible quantity is rarer under the null. It has no structural counterpart to proton parity. And, unlike 5 — whose multiples in these value sets are almost all multiples of 25, so that the larger excess one would attribute to 25, 50 or 100 only re-scores the same quantities against a smaller chance rate — the modulus 37 occurs at a single power: no quantity is divisible by 37^2, so its excess does not grow when the modulus is enlarged. Reducing each quantity to its odd part — dividing out the powers of two that the nested group hierarchy (Octet I and its sixteen- and eight-codon subgroups) introduces mechanically, since a group of size $2k$ has twice the nucleon counts of its size-k refinement — sharpens this. Among the Key 0 quantities, 37 divides ten distinct odd cores; the multiples of $73 = 2 \cdot 37 - 1$ (the value of Δ for the Octet II eight-codon groups) collapse to that single core, and the multiples of 61 to the single core $549 = 9 \cdot 61$, to which the pyrimidine proton pool $P(\text{Pyrimidine}) = 2196 = 2^2 \cdot 3^2 \cdot 61 = 4 \cdot 549$ and the proton count of each pyrimidine letter $1098 = 2 \cdot 549$ both reduce. The factor 73 is thus an echo of the modulus itself; the factor 61 is an isolated factorization unrelated to 37 (its quantities deviate from the nearest multiples of 37 by $+13$ and -12, not by ± 1 or ± 2); and the factor 5 reduces entirely to 10 and 25, every quantity here divisible by 5 being divisible by 10 or by 25. The odd-core counts are tabulated in the corresponding column of `prime_divisibility_scan.csv`.

The functional-group asymmetry across positions. One observation on the Keto/Amino axis reaches beyond the third position. The asymmetry $N(\text{Keto}) - N(\text{Amino})$ that localizes to the Trp–Ile pair at the third position (Section 4.1) equals 37 there; the same Keto/Amino axis — the $\{G, T\}$ versus $\{A, C\}$ split — yields a quantity divisible by 37 at the first position as well (N-asymmetry $-333 = -9 \cdot 37$) and at the second (T-asymmetry $185 = 5 \cdot 37$). The divisible quantity is N, T, N by position, fixed after inspection rather than in advance, so this is a single observation along one axis rather than a control of the kind above. The three are arithmetically independent: each is a sum over a different partition of the same sixty codons, and the divisibility at one position neither follows from nor implies it at another; what they share is only the choice of axis. Across the 27 NCBI translation tables this three-position coincidence is specific to the standard code and its exact codon-to-amino-acid equivalent (Table 11): no other table reaches more than one position, and the tables that preserve the Trp–Ile localization (Tables 6, 29 and 30) reproduce only the third-position value. Three caveats further qualify it. Unlike the third-position case, the first- and second-position asymmetries do not localize to a single amino-acid pair — each receives contributions from nineteen of the twenty amino acids, because the cancellation that isolates Trp and Ile relies on third-position synonymy, which the other positions lack. Their non-divisible components carry no analogue of the parity-induced residue -1 that accompanies the third-position value. And, as in Section 4.1, the 27 tables are not statistically independent, so this is evidence of specificity among natural codes rather than of improbability under a null model. The per-table values are tabulated in `position_asymmetry_ncbi.csv`.

All three controls are deterministic and exact-integer, and are regenerated by `controls.py` (Section 2.6).

4.3 Significance under a random-code null

The controls of Section 4.2 and the comparison of Section 4.1 establish that the structure is specific — to the third codon position, to the modulus 37, and, among the 27 natural codes, to the standard code — but specificity among realized codes is not improbability under an explicit null. We therefore test the standard code against a random-code null of the kind used in studies of genetic-code optimality [1]: the architecture of the code is held fixed (the 64 codons, the synonymous blocks with their sizes, the three stop codons, the Rumer partition, and the third-position axes), and the only thing randomized is which amino acid, carrying its fixed proton and neutron counts, occupies each of the twenty blocks. A trial is a uniformly random permutation of the twenty amino acids over the twenty blocks; the modulus 37 is taken a priori from the prior literature, and the test statistics are fixed in advance. The computation is reported in `mc_significance.py` and summarized in `mc_significance.csv`; the figures below are from 10^6 trials at a fixed seed. An independent re-implementation that shares no code with `mc_significance.py` — using a different construction of the codon groups and a different random-number generator — reproduces these figures (`verify_mc.py`).

Four statistics were pre-specified. First, two independent 37-anchors — the whole-pool neutron count $N(\text{All sense}) = 3589 = 97 \cdot 37$ and the Octet I total $T(\text{Octet I}) = 3700 = 100 \cdot 37$ — are both divisible by 37 in a fraction 0.0008 of random codes, against an independent reference $(1/37)^2 = 0.00073$. Second, among the four nucleon quantities of the nine groups of the Rumer/axis partition — the full pool, the six chemical-axis half-pools, and the two octets — the standard code has 13 divisible by 37, against a null mean of 1.0; a random code reaches 13 or more with frequency 4×10^{-5}. Third, restricted to the parametrization-independent groups alone, on which 37 can never be imposed, the standard code has 4 such quantities against a null mean of 1.9, a value reached or exceeded by 0.15 of random codes. Fourth, a statistic measuring not how many quantities fall exactly on the lattice of step 37 but how close the whole set lies to it: for each of the 33 groups under both keys the distance of every nonzero T, P, N, Δ to the nearest multiple of 37 is averaged (264 values). The standard code gives mean distance 5.10 against a null mean of 9.07, a value reached or bettered by 8×10^{-4} of random codes. This is not a restatement of the divisibility counts: excluding the exact multiples, the remaining quantities still lie at mean distance 7.0 from the lattice against 9.5 for a uniform residue. The concentration is stronger under Key 1 (mean distance 4.64) than under Key 0 (5.56); it is the aggregate form of the cascade of ± 1 and ± 2 deviations described group by group in Section 3.2. The comparison between the two keys is, however, not symmetric under the null: the standard code is evaluated under the Key 1 derived for it, whereas each random code is evaluated under the same fixed service-codon values $(37, 37)$ and $(1, 0)$ rather than under a key re-derived from its own balance and divisibility conditions. The Key 1 contribution therefore cannot be read as a parametrization-free gain. Restricting the statistic to Key 0 alone — where no divisibility by 37 is imposed — the standard code still attains a mean distance of 5.56 against a null mean of 9.02, reached or bettered by a fraction 2.7×10^{-3} of random codes; the concentration is thus already present before any service-codon parametrization, and the Key 1 evaluation sharpens but does not create it.

The weight of the test rests on the second and fourth statistics rather than on the two anchors of the first, which are individually weak; of these, $T(\text{Octet I}) = 3700$ is the quantity emphasized in the prior literature, while $N(\text{All sense}) = 3589$ is established here. Neither the second nor the fourth selects favorable numbers, the one counting divisibility across the entire a-priori Rumer/axis partition and the other averaging over all 264 quantities without exclusion. The first, second, and fourth statistics place the standard code far in the tail of the null; the third does not. These tail probabilities are not independent of one another, and in particular the second and the fourth are two views of the same concentration — S4 assigns distance zero to precisely the quantities S2 counts as divisible — so their p-values should not be combined multiplicatively; the joint evidence is closer to the stronger of the two than to their product. The effect therefore resides in the full sense-pool partition — in particular the divisibility by 37 of both the proton and the neutron count of the Amino and Weak groups (and hence of their totals and differences), together with the neutron counts of the full pool and of the Keto and Strong groups, the Pyrimidine Δ, and the Octet I total — rather than in the parametrization-independent subgroups by themselves. Three qualifications attach to these figures. The null fixes the standard code's degeneracy architecture and the amino-acid nucleon counts and randomizes only the assignment; a different generative model — randomizing the block sizes, or the nucleon counts — would give different numbers. The groups counted in the second statistic are the a priori Rumer and third-position chemical partition and 37 is fixed in advance, so neither was selected to fit, but the count is one of several reasonable global statistics, and its constituents are not independent. For this reason we treat the fourth statistic, which averages over all 264 quantities with no choice of which to count, as the primary measure, and the second as a corroborating illustration rather than as an independent p-value; the discrete count is the more exposed of the two to the residual freedom in choosing a global statistic. And improbability under this null concerns combinatorial codes, a different question from the specificity among the 27 natural codes of Section 4.1; it speaks to no biological mechanism.

4.4 Relation to prior work

Three prior contributions are particularly relevant to the arithmetic structure analyzed in this work. Yu. B. Rumer [3, 4] introduced the partition of the 64 codons of the standard code into two equal classes: Octet I consisting of the eight fully degenerate families, and Octet II consisting of the eight heterogeneous families. The two classes are related by the involution $R = (\mathrm{T} \leftrightarrow \mathrm{G}, \mathrm{C} \leftrightarrow \mathrm{A})$, under which each family of one class is mapped to a family of the other. Shcherbak and Makukov [5] observed that the number 37 plays a recurrent role in the genetic code, appearing as a divisor of amino acid nucleon sums across various groupings of codons. Panov and Filatov [2] revisited and extended this analysis. They introduced the per-codon counting rule, in which each of the 64 codons contributes independently the nucleon count of the amino acid it encodes. They also proposed that stop codons be assigned a total nucleon count of 74 each rather than zero.

The present work differs from these in several methodological choices. Shcherbak and Makukov, and Panov and Filatov following them, decompose each amino acid into a constant part with nucleon count 74 and a variable side chain, and many of their nucleon sums are formed over side chains alone. This decomposition forces a special treatment of proline, whose cyclic side chain reduces the nucleon count of its constant part to 73 rather than 74: to bring it in line with the other nineteen amino acids, Shcherbak and Makukov introduce a virtual transfer of one nucleon from its side chain to its constant part, an operation they call the *activation key*. The present work uses full amino acid nucleon counts without separation into constant and side-chain components, and the question of how to treat proline does not arise. Shcherbak and Makukov also adopt a counting rule in which each amino acid contributes once per codon family rather than once per codon; Panov and Filatov replace this with the per-codon rule, which the present work also uses. Finally, all nucleon sums in the prior works are formed over the total nucleon count T (or over its restriction to side chains); the separation of T into protons P and neutrons N, and the analysis of the difference $\Delta = P - N$, are introduced in the present work and underlie a class of arguments unavailable when only T is considered.

The two prior works also differ from ours in the systematics by which codons are grouped. They apply several different grouping schemes in parallel (Gamow's sorting and others), each tailored to a particular set of patterns. The present work formulates a single grouping system based on the Rumer octets and the third-position chemical axes, and analyzes the arithmetic structure within it consistently. The parametrization of service codons also differs: Shcherbak and Makukov treat stop codons as contributing zero and identify ATG with Met throughout, Panov and Filatov propose in addition the assignment of total nucleon count 74 to stop codons as an alternative, while the present work considers the parametrization of service codons as part of the derivation of Key 1 (Section 3.3). Finally, Shcherbak and Makukov work extensively with the euplotid code variant (in which TGA is reassigned to Cys), which they take as a symmetrized counterpart of the standard code; Panov and Filatov treat this as a switchable alternative. The present work focuses on the standard code, and compares its arithmetic structure with the 26 alternative NCBI translation tables on equal footing (Sections 3.3.5, 4.1).

We now describe how the divisibility findings of these prior works project onto the groups defined in the present analysis. Shcherbak and Makukov, counting one molecule per distinct coding role within a family, observed that the sum of amino acid nucleon counts over Octet I is divisible by 37, with $T(\text{Octet I}) = 925 = 25 \cdot 37$, and that the same holds for Octet II under the assignment of zero to stop codons, with $T(\text{Octet II}) = 2220 = 60 \cdot 37$. Panov and Filatov, having introduced the per-codon counting rule, obtained the corresponding totals under codon-level counting: $T(\text{Octet I}) = 3700 = 100 \cdot 37$ and $T(\text{Octet II}) = 4218 = 114 \cdot 37$ (or $4440 = 120 \cdot 37$ under their alternative assignment of 74 to stop codons).

The relation between the per-coding-role total and the per-codon total is direct for Octet I, which is fully four-fold degenerate: $3700 = 4 \cdot 925$. For Octet II, whose families contain different numbers of distinct coding roles, the per-codon total 4218 is not a multiple of the per-coding-role total 2220 and is an independent fact. The per-coding-role total 2220 itself coincides with $T(\text{Keto} \cap \text{Octet II}) = T(\text{Strong} \cap \text{Octet II})$ in our analysis: the universal pyrimidine synonymy within Octet II (Section 3.1.2) implies that the sixteen codons of Keto (or of Strong) intersected with Octet II contain each coding role exactly once, so their per-codon total equals the per-coding-role total.

From these primary facts, the divisibility by 37 of several further groups in our systematics follows trivially. The total over the entire code is the sum of the two octet totals; the sixteen-codon and eight-codon subgroups of Octet I are mutually identical by four-fold degeneracy; and the four chemical-axis groups Keto, Strong, Amino, Weak each decompose into an Octet I sixteen-codon subgroup and an Octet II sixteen-codon subgroup, of which the first is automatically divisible by 37 and the second reduces to a separate question addressed in the present work.

4.5 Limitations and outlook

The statistical significance of the observed arithmetic regularities is assessed in Section 4.3 under one explicit null — the random-code permutation, which fixes the code's degeneracy architecture and the amino-acid nucleon counts and randomizes only the assignment of amino acids to codon families. Under that null the full Rumer/axis structure is highly atypical, while the parametrization-independent subgroups alone are not. The choice of null is not unique: alternative generative models, for example randomizing the block sizes themselves or the nucleon counts, would probe different aspects of the regularity and are left for future work, as is any account of a mechanism that could produce it.

The parametrization of service codons in the present work is restricted to Key 0 and Key 1; additional parametric assignments are tabulated in `key_parameters.csv` and in the accompanying `visualization.html` for illustration of other choices.

A further direction concerns the group-theoretic structure of the codon table. Following Panov and Filatov [2], to each of the 24 orderings of the four bases on the axes of the codon matrix we associate its *calligram*: the geometric pattern formed by the eight homogeneous codon families of Octet I on the 4×4 matrix. Connectedness of an octet refers to the connectedness of this pattern under 4-adjacency of matrix cells. Among the 24 calligrams, Panov and Filatov [2] identified four — CTGA, CGTA, ATGC, and AGTC — in which each of the Rumer octets forms a connected region, and showed that these four form a closed set under the transformations $R = (\text{T} \leftrightarrow \text{G}, \text{C} \leftrightarrow \text{A})$, $R_1 = (\text{T} \leftrightarrow \text{G})$, and $R_2 = (\text{C} \leftrightarrow \text{A})$. These three transformations together with the identity form the Klein four-group V_4, and the action of V_4 extends naturally to the full set of 24 calligrams. This action of V_4 on the full set of 24 calligrams is free, partitioning it into six orbits of four calligrams each. Exactly one of these six orbits consists of connected calligrams, accounting for the four calligrams above; the remaining five orbits contain no connected calligram. This connected orbit furthermore contains the canonical chemical ordering C, G, T, A of Rumer [4] (Section 2.2); the correspondence between this chemical ordering and the geometric property of connectedness is non-automatic, since only two of the eight chemically alternating orderings of the four bases yield connected octets. The search for further structural invariants distinguishing the orbits — and for connections between the group-theoretic orbits and the arithmetic systematics of the present work — is left to future investigation.

5 Conclusion

The standard genetic code exhibits arithmetic regularities visible at several levels of parametric specificity. A first layer is independent of any choice of nucleon contributions for the service codons. The eight families of Octet I carry total nucleon count $T(\text{Octet I}) = 3700 = 100 \cdot 37$, with proton and neutron contributions $P(\text{Octet I}) = 2000$ and $N(\text{Octet I}) = 1700$. The universal pyrimidine synonymy within Octet II produces identities between the twin sixteen-codon groups Keto/Strong and Amino/Weak within Octet II.

A second layer arises under the baseline parametrization, where the proton and neutron counts P and N are analyzed separately. The total nucleon count T is divisible by 37 for the four chemical-axis groups and for Octet II. For the Amino and Weak groups, P and N are individually divisible by 37, aligning these groups with the lattice of step 37 along both axes.

Beyond these layers, an analytical derivation (Section 3.3) shows that the parametrization of service codons compatible with the arithmetic of the standard code is uniquely determined as the minimal non-negative integer solution of a system of constraints relating P, N, and the chemical-axis groups defined by the third position. The derivation rests on a sequence of structural preconditions that hold simultaneously for the standard code: the parametrization-independent values of T, P, and N over Octet I, the divisibility of N over the non-service codons by 37, the localization of the Keto/Amino asymmetry in the single pair Trp/Ile, the realization of the nucleon differences $\delta N = 37$ and $|\delta P| = 36$ by a single nucleon profile among all 190 pairs of canonical amino acids, the parity theorem on proton counts that excludes $|\delta P| = 37$ for any pair of canonical amino acids, the specific distribution of service codons across the chemical half-pools that gives rise to a deficit of exactly $111 = 3 \cdot 37$ between the Keto and Amino sense pools, and the coprimality $\gcd(4, 37) = 1$ ensuring an integer solution for the stop codon parameter. None of these preconditions is a parametric assumption; each is an empirical property of the standard code, and the analytical derivation exists because all of them hold simultaneously.

Under the derived parametrization, the arithmetic regularities organize into a connected family: alignment of the four chemical-axis groups along the axes T, P, N, and Δ (Section 3.4.1), macroscopic identities including $N(\text{All}) = T(\text{Octet I}) = 3700$ and $P(\text{All}) = T(\text{Octet II}) = 4292$ (Section 3.4.2), the uniform invariant $\Delta = 296$ across all six 32-codon groups (Section 3.4.4), and divisibility patterns by 37 and 999 at multiple levels of partitioning (Section 3.4.3). The comparative analysis across the 27 NCBI translation tables (Section 3.3.5) shows that the simultaneous holding of all structural preconditions enumerated above is a property of the standard code alone.

Two features of this structure constrain how it may be read. First, the regularities reside in group sums rather than in individual molecules: no single canonical amino acid has a nucleon count divisible by 37, and in groups of variable coding degeneracy, such as Amino, the divisibility holds only when codons are counted with their multiplicity and fails when each amino acid is counted once. The nucleon composition and the degeneracy of coding are therefore not independent quantities within this structure: the regularity is a joint property of how nucleon mass is distributed and how many codons each amino acid receives. Second, the entire first layer above, including the values over Octet I and the twin-group identities, is established with the service codons carrying no nucleon values at all; the derived parametrization of the service codons enters only at the later layers and does not affect the parameter-independent results. The assigned values are thus a device for exhibiting the arithmetic of the complete table, not a foundation on which the principal findings rest.

The specificity established by the comparison across natural codes is complemented by deterministic controls and by a statistical test. The controls show that the divisibility by 37 is confined to the third codon position and that 37 is set apart from the other small primes. Under an explicit random-code null that fixes the degeneracy architecture and the amino-acid nucleon counts and randomizes only the assignment of amino acids to codon families, the full Rumer/axis structure yields empirical p-values on the order of 10^{-4} to 10^{-3}, while the parametrization-independent subgroups alone are not atypical. This is improbability under one combinatorial null, distinct from the specificity among the 27 natural codes, and it speaks to no biological mechanism; alternative null models are left for future work.

The methodological choice underlying the present work is to restrict attention to parameters whose definition is unambiguous — the total nucleon count T, its decomposition into protons P and neutrons N, and the difference $\Delta = P - N$ — and to a single grouping system based on the Rumer octets and the third-position chemical axes. The interpretation of the arithmetic facts collected here is left open.

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